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# Reshaping electronic word-of-mouth and commitment to second-hand high-end fashion with blockchain-enabled sustainability information: Comparing Vietnam and France

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ABSTRACT

This study investigates how blockchain-enabled sustainability information factors enhance consumer-related outcomes in the second-hand high-end fashion market, including blockchain-enabled electronic word-of-mouth (BE-EWOM), customer trust (CT), perceived sustainability value (PSV), purchase intention (PI), and relationship commitment (RC). An online survey collected 465 valid responses from France and Vietnam, analyzed using PLS-SEM in SmartPLS 4. The findings indicate that the immutability of blockchain-enabled sustainability information (BES-II) and the transparency of certification information (BES-CIT) significantly enhance CT and PSV in both markets. Moreover, blockchain-enabled sustainability contribution comparison information clarity (BES-CCIC) positively impacts BE-EWOM, CT, and PSV. This effect is particularly evident in Vietnam, where it further fosters consumer commitment and trust. While BE-EWOM and PSV strongly influence consumer behavior in Vietnam, CT and RC are more critical in France. Conceptually, this study extends Affordance Theory and Trust-Commitment Theory within the context of blockchain-enabled sustainability. These findings provide practical recommendations for enterprises, regulators, and stakeholders who advocate Sustainable Development Goals (SDGs) at the intersection of emerging technology and the fashion industry. This study represents the first cross-national attempt to explain how such information shapes consumer choices in the second-hand high-end fashion sector.

## 1. Introduction

Luxury fashion is under increasing scrutiny due to its significant environmental impact, with unsustainable production practices contributing to large-scale waste and pollution (Aponte et al., 2024). In response, luxury second-hand fashion has gained traction as a viable solution aligned with circular economy principles, extending product life cycles and reducing environmental impact (Chen and Tabata, 2024). Empirical research further supports this shift, demonstrating that second-hand fashion particularly in the luxury handbag market offers the most ecological benefits compared to both traditional production and plant-based alternatives in terms of climate impact, water depletion, and human toxicity (Shou and Domenech, 2022).

However, the adoption of online second-hand luxury fashion remains

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constrained by authenticity risk and information asymmetry. These frictions remain hindered by issues of trust, transparency, and perceived value (Frank et al., 2023; Xiao et al., 2024). These challenges are especially pronounced in the high-end second-hand segment, where products such as Hermès or Chanel items carry high financial value and symbolic meaning but are highly vulnerable to counterfeiting. Unlike first-hand purchases made through official channels, second-hand transactions are often conducted via digital platforms where the absence of institutional oversight renders authentication and credible sustainability information critical for building consumer trust. Consumers seeking to purchase second-hand luxury goods frequently encounter challenges in verifying product authenticity, as documentation such as invoices or certificates can be forged or reported lost. Consequently, trust in this market is often based on the perceived reputation of sellers or platforms, rather than on verifiable, objective guarantees. This reliance on informal signals underscores a key concern.

Blockchain is increasingly proposed as a verification infrastructure in resale settings. As a distributed ledger technology, it can enable verifiable authenticity checks and credible sustainability disclosure, reducing information asymmetry in high-risk online second-hand luxury transactions and supporting secure verification of product authenticity and sustainability certifications (Groβe et al., 2024; Hina et al., 2024a,b; Tripathi et al., 2021; Ho et al., 2024). More broadly, such verifiability and transparency can strengthen trust-relevant cues in digital commerce contexts (Isabel et al., 2024; Shin and Hwang, 2020). Related work further suggests that blockchain immutability and transparency can increase the reliability of user ratings, contributing to stronger customer trust (CT) and more authentic interaction (Singh and Sharma, 2022; Panghal et al., 2023; Le, 2024; Duan and Zhu, 2024).

However, blockchain research has primarily examined transparency and traceability in upstream supply chains, including fashion-related sourcing and logistics, rather than consumer-facing decision-making in second-hand luxury resale (Gurtu and Johny, 2019; Sunmola and Burgess, 2023; Kamilaris et al., 2019; Isabel et al., 2024). As a result, there is still relatively little research on how blockchain-enabled information shapes consumer psychology in second-hand luxury fashion and through which mechanisms such as CT, perceived sustainability value (PSV), and blockchain-enabled electronic word-of-mouth (BE-EWOM), these perceptions translate into relationship and purchase outcomes (Sin Ying and Mok, 2023; Swazan and Youn, 2024). This limitation is consequential because emerging technologies integrated across the fashion consumption cycle may mitigate pollution impacts, yet remain under-examined in consumer-facing resale settings (Schiaroli et al., 2024). Recent scholarly dialogue has therefore called for studies that test whether, and through what mechanisms, blockchain benefits CT and purchase decisions alongside other credibility tools such as third-party certification or quantified sustainability claims (Mathew and Spinelli, 2025).

This research enhances the current understanding by addressing four key gaps in the existing literature. First, while blockchain has been well studied in supply chain management, its influence on CT and purchase intention (PI) in second-hand fashion is an underinvestigated area (Jiang et al., 2023; Schiaroli et al., 2024; Swazan and Youn, 2024). Second, while electronic word-of-mouth (EWOM) plays a crucial role in fashion-related PI, studies on blockchain-enabled EWOM (BE-EWOM) and its impact on sustainable fashion consumers are still scarce (Boardman et al., 2022; Schiaroli et al., 2024). Authenticity concerns are especially pronounced in second-hand fashion transactions (Safeer et al., 2021). Third, despite the various features of blockchain such as transparency, immutability, and traceability, it remains unclear which of these features most effectively enhance CT in platforms adopting blockchain, particularly in the second-hand luxury fashion market (Groβe et al., 2024; Mathew and Spinelli, 2025). Previous research has mainly focused on the application of blockchain in food supply chains and logistics (Treiblmaier and Garaus, 2023; Ho et al., 2024; Duong et al., 2025), with little discussion of how its role has changed consumer behavior when applied in second-hand fashion markets. Also, there is a paucity of comparative studies examining distinct types of blockchain adoption across different fashion markets (Chen, 2023; Schiaroli et al., 2024). France is a world leader in luxury clothing, whereas Vietnam represents a rapidly digitalising market, offering a meaningful contrast for examining blockchain integration in resale settings (Bao and Phan, 2022). Drawing on Hofstede as a baseline, consumers may differ in how they evaluate authenticity and sustainability cues across these contexts, for example, higher uncertainty avoidance may be associated with stronger reliance on established brand heritage and formal safeguards in credibility judgments (Hofstede, 2001; Boissieu and Chaney, 2024). Vietnamese consumers, characterized by high collectivism, rapid digital adoption, and less standardised institutional safeguards in second-hand transactions, may face a trust deficit in second-hand transactions, making them more likely to anchor trust in technological affordances such as platform transparency and digital verification (OECD, 2023; Nguyen and Truong, 2016). Taken together, these contextual contrasts suggest that blockchain-enabled sustainability information may activate different trust-formation pathways and sustainable value evaluations across the two markets. Yet, despite this theoretical rationale, existing research has not provided a comparative account of how blockchain enhances CT and PI in culturally and institutionally distinct second-hand luxury fashion contexts.

Building on this rationale, the present study applies affordance theory to examine how specific features of blockchain are interpreted and acted upon differently in France and Vietnam. Focusing on the three key affordances of immutability, transparency, and clarity, the study investigates how blockchain-enabled information embedded in second-hand luxury fashion platforms shapes CT, perceived sustainability value (PSV), EWOM, relationship commitment (RC), and PI. The findings provide benefits for brands, e-commerce platforms as well as governments, supporting the development of blockchain-driven sustainable fashion initiatives and encouraging more responsible consumption within the fashion industry. This study adopts a quantitative approach using cross-national survey data and structural equation modeling to empirically test these relationships.

Taken together, these gaps in the literature indicate that existing research has not yet clarified (1) which blockchain affordances are most influential in shaping CT, PSV, and EWOM in second-hand luxury fashion platforms, (2) how these consumer perceptions translate into RC and PI, and (3) whether these mechanisms operate differently across national contexts. As prior studies typically examine blockchain features, trust, or sustainability perceptions in isolation, there remains a limited integrative understanding of how blockchain-enabled sustainability information simultaneously shapes cognitive (PSV), relational (CT, RC), and behavioral (EWOM, PI) outcomes. Addressing these unresolved issues requires a theoretically grounded and cross-national investigation, which directly motivates the research questions proposed in this study. Accordingly, the study addresses the following research questions:

- *RQ1: How do the affordances of blockchain (transparency, immutability, and clarity) influence customer trust, blockchain-enabled electronic word-of-mouth, and perceived sustainability value in the second-hand luxury fashion market?*
- *RQ2: What role do blockchain-enabled electronic word-of-mouth and perceived sustainability value play in strengthening customer trust?*
- *RQ3: How do customer trust, blockchain-enabled electronic word-of-mouth, and perceived sustainability value affect relationship commitment and purchase intention?*
- *RQ4: How do customer trust, blockchain-enabled electronic word-of-mouth, and perceived sustainability value, relationship commitment, and purchase intention differ between French and Vietnamese customers when blockchain is integrated into second-hand luxury fashion platforms?*

## 2. Literature review

### 2.1. Theoretical background

#### 2.1.1. Theory of affordance and blockchain affordances

The concept of Affordance discusses that an environment offers to an organism its action possibilities, reflecting a certain relation of an organism and the objects or features in its environment showing what interactions are possible (Gibson, 2014/2014). Following this, the affordance has been expanded into the technology context to describe communications between users and digital interfaces that affect user experience (Evans et al., 2017; Tang and Zhang, 2020; Volkoff and Strong, 2017). This relational perspective is particularly relevant for technologies like blockchain, whose value arises not merely from its technical features but from how users perceive its capacity to support trustworthy, interpretable, and verifiable information exchanges. In this sense, blockchain's technical characteristics can be viewed as enabling conditions, while affordances emerge through users' interaction with blockchain-enabled information as presented via platform interfaces and disclosure mechanisms (e.g., product passport pages, QR/NFC-enabled records, or on-platform verification dashboards).

Blockchain technologies have been associated with several affordances, including transparency, traceability, immutability that reshape how information is recorded, accessed, and evaluated (Du et al., 2019; Szabó et al., 2022). Importantly, these affordances are realised when consumers recognize and utilize action possibilities (e.g., inspecting provenance and sustainability claims, comparing alternatives, or sharing verification cues), rather than residing solely in the technical properties themselves. Accordingly, in line with affordance theory, we conceptualize blockchain affordances as perceived action possibilities - consumers' perceived ability to access, verify, and use blockchain-enabled information for evaluation and decision-making - rather than as technical features per se.

This study focuses on the three affordances most directly linked to consumer-facing sustainability information: transparency, immutability, and clarity. Transparency affords direct visibility and verifiability, enabling consumers to inspect authenticated product claims without reliance on institutional intermediaries (Shin and Hwang, 2020; Shrestha et al., 2021; Choi et al., 2021). When consumers perceive they can independently verify information, the affordance reduces information asymmetry, an issue particularly salient in the luxury second-hand market, where provenance uncertainty is common. Immutability affords permanence and tamper-resistance, allowing consumers to rely on the stability of sustainability records stored on-chain (Politou et al., 2021; Hina et al., 2024a,b). This affordance directly supports perceptions of integrity, as it signals that data cannot be altered retroactively, thereby strengthening trust in claims about product origin, ownership, or environmental impact (Jahanbin et al., 2023). Clarity, by contrast, is conceptualized in this study as a cognitive affordance that makes blockchain-enabled sustainability information not only easier to understand but also easier to compare across products in ways that support purchase decisions. Whereas transparency affords visibility into the underlying data and immutability affords confidence in its stability, neither guarantees that non-expert consumers can interpret complex sustainability metrics or use them to differentiate among alternatives. Clarity addresses this gap by capturing consumers' perception that blockchain-based sustainability information is structured, intelligible and decision-useful, thereby reducing cognitive burden and ambiguity in sustainability evaluations (Hina et al., 2024a,b; Hina et al., 2024a,b; Liu et al., 2023). Prior work on information disclosure shows that transparency enhances trust and behavioural responses only when the disclosed information is comprehensible and interpretable (Lam et al., 2020). Accordingly, clarity is treated here as a distinct cognitive affordance that conditions whether blockchain-enabled sustainability information becomes actionable in consumer decision-making. Traceability, although widely discussed in blockchain literature as the ability to follow the chain of custody across stages, is not operationalised in the present model; instead, we focus on transparency, immutability, and clarity as the affordances most directly linked to consumer-facing sustainability information in platform-mediated purchase decisions (Jahanbin et al., 2023; Duong et al., 2024).

Previous research has explored the application of affordance theory in diverse areas, such as mobile apps, social platforms, and gaming (Gholizadeh et al., 2021; Wang et al., 2022; Sun et al., 2023), highlighting its relevance for explaining how users perceive and act upon technology-enabled information in platform settings. The literature discusses in detail blockchain's benefits to improve the immutability and improve clarity of supply chains as well as making them more transparent (Shin and Hwang, 2020; Jahanbin et al., 2023). When adopted by consumers, these systems increase the visibility of sustainability information compared with more ambiguous traditional channels, empowering more informed sustainable-consumption decisions and supporting the acceptance of blockchain solutions in consumer markets (Nygaard and Silkoset, 2022; Cozzio et al., 2023; Clohessy and Acton, 2019). Table 1 summarises studies relevant to blockchain affordances and consumer-related outcomes.

Importantly, in consumer-facing resale platforms, blockchain-enabled sustainability information is delivered through retail service interfaces and governance mechanisms. Prior e-commerce and service research shows that institutional structures such as authentication services, guarantees, return policies, and dispute resolution, are central to trust formation under uncertainty (McKnight et al., 2002; Pavlou, 2003). Relatedly, perceived e-service quality and service assurance shape how consumers interpret and act upon transparency and verification cues (Parasuraman et al., 2005). Thus, blockchain affordances are more likely to translate into customer trust and downstream behaviours when embedded in credible platform services that make verification useable and consequential. Therefore, building on affordance theory (Gibson, 2014/2014) as its theoretical support and underpinned in the information systems literature (Volkoff and Strong, 2017) and earlier research on sustainable consumption (Cozzio et al., 2023; Nygaard and Silkoset, 2022), this study identifies transparency, immutability, clarity as key perceived affordances, highlighting their critical influence on customer trust, electronic word-of-mouth and perceived sustainability value in blockchain-enabled, platform-mediated retail contexts.

#### 2.1.2. The Commitment-Trust Theory

Commitment-Trust Theory suggests that success in Relationship Marketing will largely depend on the presence of trust and commitment (Morgan and Hunt, 1994). Trust refers to confidence in a partner's reliability and integrity, whereas commitment reflects an enduring intention to maintain the relationship over time. In the second-hand luxury fashion market, concerns about whether products are authentic or counterfeit can undermine consumer confidence and impede adoption. Therefore, trust plays an important role in the second-hand luxury fashion market (Boissieu et al., 2021). Online resale platforms have expanded access to second-hand luxury, enabling buyers to purchase at lower prices (Ohta, 2022). Consumers may unwittingly buy second-hand luxury items that they believe are authentic but turn out later to be fake (Liu et al., 2013). This creates a significant barrier to market growth.

Blockchain technology emerges as a potential solution, offering secure, transparent, and immutable production certification (Casino et al., 2019). By establishing unalterable transaction records, blockchain allows consumers to independently verify the authenticity and ethical sourcing of luxury goods, fostering a form of cognitive trust grounded in verifiable evidence of product integrity (Mayer et al., 1995). Furthermore, features such as smart contracts, which operate as self-executing agreements embedded directly in code, enhance transactional efficiency by reducing intermediary dependence, lowering costs, and improving transparency across the resale process (Ray et al., 2024). Nevertheless, the trust cultivated through these technological capabilities extends beyond facilitating smoother exchanges. Consistent with

**Table 1**
Summary of prior studies on blockchain affordances.

| Authors | Antecedents | Mediator/Moderator | Outcomes | Theory | Country/ Territory | Key Findings |
|---|---|---|---|---|---|---|
| Shin and Hwang (2020) | Security, Privacy, Traceability | Trust, Transparency, Reliability | Satisfaction | Affordance Theory, Expectation Confirmation Theory | Korea | Blockchain's perceived transparency and reliability significantly enhance users' trust and satisfaction, indicating that trust functions as a heuristic through which users interpret technological and affective affordances. |
| Shrestha et al. (2021) | Blockchain-enabled traceability | Perceived blockchain-related information transparency, Blockchain-based trust | Purchase intention | Technology Acceptance Model | Five continents (Africa, Asia, Europe, N. America, Oceania) | Trust and adoption intentions depend strongly on how transparently blockchain systems communicate privacy and security features, reinforcing the role of information transparency as a critical mechanism shaping user acceptance. |
| Liu et al. (2023) | Consumer blockchain technology experience, Environmental information transparency perceived | Digital trust, Swift trust | Impulse buying | Signal Theory, Trust Transfer Theory | China | Information transparency is an important mechanism inherent in blockchain technology to influence consumer trust. |
| Hina et al. (2024) | Transparency, Traceability, Immutability | Efficiency, Social impression, Trust, Sustainability information clarity | Purchase Intentions | Affordance Theory, Theory of Consumption Values | European | Blockchain affordances positively affect consumption values, including efficiency, social impression, trust, and sustainability information clarity. |
| Duong et al. (2024) | Blockchain-based food traceability system | Trust, Technology anxiety | Purchase intention | The trust theory, Stimulus-Organism-Response theory | Vietnam | Blockchain-based food traceability systems notably enhance consumer trust in organic food products and their intention to buy these products. |
| Hina et al. (2024) | Blockchain transparency, Information seeking | Trust, Information Seeking | Consumer satisfaction, Sustainable product uncertainty | Signaling theory | European | The relationship between blockchain transparency and satisfaction is reduced by information-seeking. Consumers' experience of blockchain transparency depends on their inclination to seek information. |

Morgan and Hunt's framework (1994), Geyskens et al. (1996) demonstrate that higher levels of trust can shift relational motivations away from calculative dependence and toward affective attachment, positioning trust as the primary driver of affective commitment. When trust is strong, even highly interdependent relationships become less anchored in cost–benefit reasoning and more aligned with a sense of partnership and mutual understanding (Fullerton, 2005).

However, this commitment-trust pathway is not globally uniform as it is subject to significant cross-national variation, powerfully moderated by factors such as cultural context. As Hofstede (2001) suggests, cultural dimensions can serve as key moderators of this process. In particular, the individualism collectivism and uncertainty avoidance dimensions are especially relevant for understanding how consumers extend trust from close others to impersonal, technology-enabled systems. In individualistic societies, where generalized trust (i.e., the willingness to trust strangers and impersonal systems) tends to be higher, social cooperation regularly extends beyond intimate networks (Kyriacou, 2016). In such contexts, blockchain may align seamlessly with prevailing social norms, serving as a logical extension of the institutional safeguards already ingrained in the social fabric. In such high-individualism contexts, blockchain-based assurances may be more readily accepted as legitimate proxies for institutional safeguards, reinforcing the trust commitment pathway. Conversely, in collectivistic societies, trust is often powerfully reserved for the in-group, such as family and close associates, resulting in comparatively lower levels of generalized trust (Kyriacou, 2016). Here, blockchain's purely technical guarantee may encounter initial skepticism, as it lacks the relational cues essential for local trust-building practices. Consequently, for blockchain to be effective, its technological proof may need to be strategically embedded within or endorsed by trusted relational channels, thereby bridging the gap between impersonal system-trust and deeply held relational-trust norms. Beyond Hofstede's value dimensions, contemporary perspectives emphasize that cross-national differences in technology-enabled trust are also shaped by institutional conditions and shared credibility norms. From an institutional lens, the perceived legitimacy of regulatory/certification regimes and platform governance can strengthen or weaken consumers' willingness to rely on impersonal technological proof (Scott, 2014; Krell et al., 2016). Cultural consensus approaches further suggest that communities may differ in what they regard as credible evidence, which can influence how blockchain-enabled verification cues translate into trust and, subsequently, commitment (Weller, 2007; Anders and Batchelder, 2012). Building upon this theoretical foundation, the present paper examines how blockchain adoption may shape consumer behavior in the luxury resale market by strengthening the dual pillars of trust and commitment.

To integrate this relational exchange logic with the technology-interaction lens of affordance theory, we specify how blockchain affordances become relationally meaningful in platform-mediated resale. Blockchain features become consequential when consumers perceive and enact affordances through the interface, enabling verification and interpretation that reduce uncertainty and perceived risk and build evidence-based (cognitive) trust (Gibson, 2014/2014; Volkoff and Strong, 2017; McKnight et al., 2002; Pavlou, 2003). This cognitive trust then functions as a key precursor to relationship commitment and downstream behaviours such as EWOM and purchase intention (Morgan and Hunt, 1994; Geyskens et al., 1996).

### 2.2. Hypotheses development

#### 2.2.1. Blockchain-enabled sustainability information immutability

Blockchain-enabled Sustainability Information Immutability (BES-II) is the capability through blockchain technology to ensure that sustainability information cannot be changed. In this study, immutability specifically concerns environmental and social impact records (e.g.,

carbon footprint, material provenance, labor/certification events) that are anchored on-chain and independently verifiable, thereby constraining retroactive alteration of sustainability claims. Blockchain's immutability means that all data entered into the Blockchain system cannot be edited or deleted (Politou et al., 2021). Once a block has been appended to the chain, it is close to impossible to alter or delete it, which helps preserve the integrity and reliability of this information (Landerreche and Stevens, 2018). In a sustainable area, blockchain can record and protect important data, including a business's environmental impact, how it performs its social responsibility, and various ecological quality certifications. EWOM includes user-generated content, social media comments, and online reviews, all of which can be manipulated to further malicious objectives (Mayzlin et al., 2014; Cai and Zhu, 2015). Negative reviews may be deleted by sellers or reported for removal (Hendy, 2019). This degrades the transparency of EWOM, to the detriment of consumers themselves trying to buy wisely, because they cannot easily distinguish between good collective wisdom and misinformation from a biased source. The immutability of blockchains has the kindness of being able to make this fraud difficult by ensuring that data cannot be erased or changed. Blockchain technology can block certain false ratings, such as slander or whitewashing (Cai and Zhu, 2016). In the context of this research, blockchain-enabled systems help limit seller-driven manipulation of EWOM sentiment. This is especially important in luxury second-hand goods markets, where consumers rely on EWOM to judge product authenticity and trustworthiness (Lim and Chung, 2011). When consumers recognize that blockchain renders sustainability claims tamper-proof and independently verifiable (i.e., records cannot be faked, erased, or retroactively edited), they are more likely to engage in BE-EWOM to share credible sustainability insights and authenticate product histories. Based on the above arguments, we propose the following hypothesis:

**H1a**. Blockchain-enabled Sustainability Information Immutability positively impacts Blockchain-enabled EWOM

In relationships, trust is crucial; it is the expectation that one trading party will act with good conscience and in a reliable manner toward another party (Shin, 2010). By reducing perceived risks and uncertainties, trust helps customers make more informed purchases (Gefen and Straub, 2004; Basha and Lal, 2019). By conducting a cross-sectional correlation analysis of blockchain technology, CT, and PI, we can see that blockchain strengthens CT through its reliability in ensuring that information is immutable and can never be deleted or manipulated to create green lies (Shin and Hwang, 2020). Unlike databases that can be changed retroactively, blockchain records are made up of time-stamped and cryptographically secured blocks of data that cannot be altered or deleted (Beck et al., 2018). This helps ensure the transparency and credibility of corporate sustainability claims and reduces information asymmetry. In a study on information security, the author defines information security as preventing unauthorized access, use, disclosure, disruption, alteration, or destruction of data and systems (Sattarova and Kim, 2007). In the context of the study, blockchain's immutability is quite close to this definition. Bart et al. (2005) concluded that when security-based measures provide protections in a way that places customers' trust in personal information, websites will significantly increase customer trust. On this basis, we propose the following:

**H1b**. Blockchain-enabled Sustainability Information Immutability positively impacts Customer Trust

According to Lee (2020), Perceived Sustainability refers to consumers' belief that a company's sustainability efforts effectively address present and future needs while aligning with ecological systems. The same study defines Perceived Value as the sum of consumers' estimates of a company's sustainability efforts. Perceived Sustainable Value (PSV) is the value that consumers perceive of a company's sustainability efforts based on its positive environmental and social actions. Immutable blockchain records provide customers with consistent and auditable sustainability metrics, thus reinforcing their impression of a company's commitment to social and environmental responsibility (Montecchi et al., 2019). By eliminating the risk of post-event changes, it provides both functional and perceived value. In other words, it allows customers to independently verify sustainability claims (Cozzio et al., 2023). Blockchain, by making sustainability claims tamper-proof and permanently recorded, allows consumers to have greater trust and enhances the perceived legitimacy of a company's sustainability initiatives. Based on these arguments, we propose the following hypothesis:

**H1c**. Blockchain-enabled Sustainability Information Immutability positively impacts Perceived Sustainability Value.

#### *2.2.2. Blockchain-enabled sustainability certification information transparency*

Blockchain-enabled Sustainability Certification Information Transparency (BES-CIT) refers to the use of blockchain technology to ensure authenticity, traceability, and transparency in sustainability certifications. Choi et al. (2021) discovered that blockchain is used to show transparent information, cutting risks down to size, and allowing companies to go on ethical sourcing and sustainability practices. Blockchain follows the principle of full transparency, so that even transactions that have been hashed or encrypted must still be visible to all the nodes participating in verification (Christidis and Devetsikiotis, 2016).

Blockchain prevents greenwashing by enabling consumers to verify the authenticity of sustainability certifications, which in turn promotes trust in the brand (Shin and Hwang, 2020). When information about sustainability certifications is transparently provided, the social benefits of customers can be consolidated by having access to accurate and truthful information about products and brands from other customers. According to Hennig-Thurau et al. (2004), this will promote the development of EWOM, consumers are more likely to share reliable information. This transparency not only ensures the quality of the information but also increases consumer participation in sharing positive experiences and reviews. Therefore, we propose the following hypothesis:

**H2a**. Blockchain-enabled Sustainability Certification Information Transparency positively impacts Blockchain-enabled EWOM

According to Liang et al. (2023), features of blockchain can increase trust in sustainable products, prompting more sustainable consumption decisions. BES-CIT helps to bridge the information gap between businesses and consumers, which in turn promotes CT (Behnke and Janssen, 2020). Also, according to signaling theory, transparent and verifiable information helps to lessen the information asymmetry between product providers on the one hand and consumers on the other, and then increases trust (Spence, 1978; Connelly et al., 2011). This, in turn, means that if consumers can independently verify sustainability information in a clear and transparent way, they will trust it all the more. In this study, blockchain makes sustainability certification information transparent. Thus, customers are able to independently verify sustainability claims free of charge, lessen skepticism over certification accuracy, and have more trust in sustainable products. Hence, the hypothesis:

**H2b**. Blockchain-enabled Sustainability Certification Information Transparency positively impacts Customer Trust

Regarding perceived sustainability value, Liang et al. (2023) believe that blockchain technology can provide consumers with independently verified information to help them make more informed choices about sustainable consumption. Blockchain transparency influences the reliability of sustainable products and enhances customer satisfaction, assisting in removing consumer ambiguity regarding sustainable products, according to a study by Hina et al. (2024a,b). When a product's sustainability can be trusted and the transparency this brings encourages consumer satisfaction, then sustainability value is seen to be good. As a result, consumers look at purchases of products that have won this trust with more favorable perceptual attitudes towards their own satisfaction of ecological needs. In summary, by making sustainability certification

transparent, blockchain allows consumers to independently verify claims, reduces skepticism about certification accuracy, and strengthens the perceived sustainability value of the product. On this basis, the following hypotheses are put forward:

**H2c**. Blockchain-enabled Sustainability Certification Information Transparency positively impacts Perceived Sustainability Value

#### *2.2.3. Blockchain-enabled sustainability contribution comparison information clarity*

Blockchain-enabled Sustainability Contribution Comparison Information Clarity (BES-CCIC) is the extent to which blockchain technology facilitates a clearer, more understandable, and easily comparable way-of presenting sustainability information. Rather than adding more data, BES-CCIC captures how blockchain helps consumers access and interpret certified information about the environmental and social impacts of products, thereby supporting informed decision-making and confidence in sustainability claims (Saberi et al., 2019; Kamble et al., 2021). Alternatively, it may be accepted as a substitute for something that becomes more common across different sources of information (Giannakis and Papadopoulos, 2016). Furthermore, consumer decisions can be significantly enhanced by having a precise and comprehensive understanding of the sustainability information (Hina et al., 2024a,b). The rise of information diffusion via EWOM, further amplifies this effect (Cheung and Thadani, 2010; See-To and Ho, 2014).

**H3a**. Blockchain-enabled Sustainability Contribution Comparison Information Clarity positively impacts Blockchain-enabled EWOM

BES-CCIC is crucial for gaining customer trust. By providing data that can be verified, blockchain means that consumers now have the opportunity to decide upon eco-friendly products (Hina et al., 2024a,b). It strengthens the relationship between supplier and consumer by reducing risk in sustainable consumption (Hina et al., 2024a,b). Safe sustainability data, ensured by blockchain, can help overcome asymmetry in information. Thus, consumers are able to gauge the credibility of sustainability claims (Saberi et al., 2019), and are more likely to trust in brands that are clear about environmental product information (Barlas et al., 2023).

**H3b**. Blockchain-enabled Sustainability Contribution Comparison Information Clarity positively impacts Customer Trust

When customers have information that can be compared about the sustainability of different goods and services, they are in a better position to determine the value of sustainable choices (Zeithaml, 1988). Going forward, Blockchain technology has the potential to significantly increase perceptions of these other values by simply showing easy to understand information on the social and environmental benefits of making sustainable choices (Hina et al., 2024a,b). As a result of these clarities, the added value of sustainable consumption can better be appreciated (Rajeev et al., 2017). Hence, BES-CCIC increases the perceived value of sustainable consumption by closing the gap between sustainability efforts and consumer understanding.

**H3c**. Blockchain-enabled Sustainability Contribution Comparison Information Clarity positively impacts Perceived Sustainability Value

#### *2.2.4. Blockchain-enabled EWOM*

EWOM refers to consumers' positive or negative statements about products, brands, or companies that are shared through digital platforms (Cheung and Lee, 2012; Hennig-Thurau et al., 2004; Chetioui et al., 2020). According to Babić Rosario et al. (2020), EWOM represents a form of user-generated communication that conveys personal consumption experiences, primarily intended for other consumers via digital platforms. However, traditional EWOM is highly vulnerable to fake or manipulated reviews due to anonymity, platform-driven biases, and the ease with which unverified users can generate misleading content (Bulchand-Gidumal and Melián-González, 2023). Recent research suggests that emerging technologies such as blockchain can strengthen the credibility and diagnostic value of consumer-generated reviews by enabling immutable, transparent, and verifiable information trails (Bulchand-Gidumal and Melián-González, 2023; Nguyen et al., 2025). Building on this perspective, we conceptualize Blockchain-Enabled Electronic Word-of-Mouth (BE-EWOM) as a technologically enhanced form of EWOM in which consumers' experiences, opinions, and recommendations explicitly anchored in blockchain-verified information. Through the features of immutability and traceability, BE-EWOM becomes tamper-resistant and fully auditable, thereby enhancing the transparency, credibility, and reliability of the shared content. For instance, a consumer purchasing a second-hand Chanel handbag on a luxury resale platform may share an online review that includes blockchain-verified proof of authenticity, such as immutable records of ownership transfers or material sourcing. Relationship commitment describes a consumer's long-term intention to continue engaging with a brand that they find valuable and mutually rewarding (Morgan and Hunt, 1994). Traditional EWOM channels have long contributed to this bond by enabling individuals to confirm each other's beliefs about a brand's authenticity, quality, and shared values (Busalim et al., 2022). With blockchain in the picture, these conversations gain even more credibility, as consumers can refer to immutable records, such as proof of ethical sourcing, fair labor, or sustainability certifications, to support what they say (Swazan and Youn, 2024). Transparency enhances consumers' trust that a brand is delivering on its promises. When these verifiable claims are echoed and reinforced through peer interactions, it can strengthen emotional attachment and, in turn, lead to greater commitment in the brand relationship. Over time, these strengthened connections cultivate a sense of community and loyalty, as consumers view the brand and its BE-EWOM network as reliable partners in their shared ethical or environmental aspirations (Hina et al., 2024a,b). Therefore, we hypothesize:

**H4a**. Blockchain-enabled EWOM positively affects relationship commitment.

PI often hinges on the perceived risk of product claims, especially in sustainability or premium sectors, where skepticism about greenwashing or overhyped benefits runs high (Hina et al., 2024a,b). EWOM traditionally addresses this gap by connecting consumers with peer experiences, but it can still leave questions about the veracity of those opinions. In contrast, BE-EWOM allows user endorsements and critiques to be supported by tamper-proof, on-chain data, such as carbon footprint metrics or raw material sourcing information. This verifiable transparency helps bridge the gap between consumer interest and purchasing action, thereby exerting a more direct and credible influence on buying decisions (Treiblmaier and Garaus, 2023). Because blockchain records are tamper-proof and verifiable, they help support what users say in their reviews. This makes peer opinions feel more trustworthy, lowers the sense of risk for buyers, and helps build confidence in what the product claims to offer (Shin and Hwang, 2020). This matters even more in high-risk or luxury markets, where earning customer trust plays a crucial role in influencing PI (AL-Issa et al., 2022; Jain et al., 2022). Therefore, hypothesis was suggested:

**H4b**. Blockchain-enabled EWOM positively affects purchase intention.

Trust has long been recognized as a foundation of brand loyalty (Morgan and Hunt, 1994), shaping whether consumers feel comfortable supporting or recommending a brand. Although consumer-generated EWOM often carries more weight than brand-created content (Babić Rosario et al., 2020), its credibility can still be questioned, especially in cases of fake reviews or vague, unverified claims (Negash and Akhbar, 2024). When blockchain is integrated into the EWOM process, it gives consumers added confidence by linking reviews to verifiable product information. This helps ease concerns about greenwashing or misinformation (Shin and Hwang, 2020). The combination of peer opinions and blockchain-verified data creates a more transparent and trustworthy environment. Over time, this kind of reinforced credibility doesn't just build trust—it also encourages consumers to stay loyal and speak

positively about the brand to others (Hina et al., 2024a,b). In turn, that creates a feedback loop of trust and advocacy, helping strengthen the brand's position in the long run. Based on this, we propose the following hypothesis:

**H4c**. Blockchain-enabled EWOM positively influences Customer Trust.

### 2.2.5. Customer trust

According to Morgan and Hunt (1994), CT is essential for online environments, where long-term relationships flourish. This emotional link will attract customer loyalty, as familiarity often breeds liking among people who are confident they will not be let down by a brand (Saleem et al., 2019). In this study, trust refers to CT in blockchain-enabled platforms-where authenticity, sustainability, and transparency are delivered through blockchain integration-rather than in the technology or brands alone (Groβe et al., 2024). When customers perceive such platforms as trustworthy, they feel secure in the truthfulness of the information supplied. This reflects the level of commitment these relationships have from the clients' side (Ramanathan et al., 2017). The role of trust is a key factor in shaping consumer behavior and influencing PI in online transactions (Gefen and Straub, 2004). When the level of trust is high, the perception of risk and uncertainty of an event is reduced, which can help consumers make more confident purchase decisions (Basha and Lal, 2019). In the case of pre-fashion products driven by trust, backed by blockchain, trust in authenticity and sustainability has increased cognitive value, driving purchase intent (Duong et al., 2024). As a result, CT significantly increases purchase intent. This is especially true in the luxury pre-fashion market, where transparency and sustainability are increasingly important (Hajli et al., 2017; Meira et al., 2024). Trust is a reliable predictor of purchase intent (Nguyen and Nguyen, 2021). CT fully affects a person's PI, particularly in online transactions (Yeon et al., 2019; Zhao et al., 2019). In the luxury second-hand fashion market, where authenticity and sustainability are primary concerns, there has been very little research on the role of CT in influencing purchase behavior based on blockchain technology. Differences in technology readiness levels between Vietnam and France may further affect the impacts of trust on PI. Thus, this investigation looks at the role of CT in their purchase intention within blockchain-enabled spaces across two different markets.

**H5a**. Customer Trust positively impacts Purchase Intention.

CT is a vital part of the local effect on relationship commitment and brand in the luxury second-hand fashion market for these two countries, Vietnam and France (Javed and Wu, 2020; Jiang et al., 2019). Nevertheless, until now, no research has been done on how CT affects relationship commitment in this blockchain-enabled second-hand clothing market. It should also be noted that there is a lack of research into differences between Vietnamese and French consumers 'behaviors at a deep level. Given these reasons, such an exploration would provide better angles to understand distinctions.

**H5b**. Customer Trust positively impacts Relationship Commitment.

### 2.2.6. Perceived sustainability value

For an extended period, the luxury sector remained largely overlooked by critics due to its relatively small size. However, it has increasingly garnered attention from non-governmental organizations and social commentators (Bendell and Kleanthous, 2007). In the modern era, consumers are increasingly considering factors beyond traditional values when evaluating luxury products (Atkinson and Kang, 2021). As a result, many companies in the luxury fashion industry have taken their efforts to display sustainable scenes through social media, receiving strong backing from online communities. Notable initiatives can be listed out such as Gucci's "Equilibrium" program, which illustrates environmental responsibility and the brand's commitment to sustainability or Burberry's "Our Planet" campaign, which showcases the adoption of eco-friendly materials and efforts toward carbon-neutral production. The researchers utilize affordance theory, which emphasizes the potential actions emerging from the interaction between an agent and its environment, as a way to explain this shift in consumer behavior. In previous research, the affordance theory has been taken as the basis for various contexts, including social media, mobile applications, and consumer decision-making (Wang et al., 2022; Gholizadeh et al., 2021; Sun et al., 2023). Within the realm of blockchain technology, affordance theory has effectively explained how transparency, traceability, and immutability optimize supply chain operations (Shin and Hwang, 2020; Jahanbin et al., 2023), that enables consumers to access up-to-date product information, authenticate product origins, and verify the integrity of data records (Cozzio et al., 2023; Nygaard and Silkoset, 2022). Based on that, the increasing significance of consumers evaluating product sustainability in building trust, especially in the decision-making process, has been highlighted. To expand upon this basis, we propose the following hypothesis:

**H6a**. Perceived sustainability value positively impacts Customer Trust

We introduce two further hypotheses that dive deeper into the interplay between trust and relationship dynamics in order to build upon the prior hypothesis suggesting that CT is enhanced by PSV. According to Morgan and Hunt (1994), trust and commitment are closely related in an exchange relationship. When a brand's sustainable efforts are believed to be genuine and meaningful (Morhart et al., 2015), consumer's trust in the brand increases, which can lead to a stronger and more enduring commitment (Chaudhuri and Holbrook, 2001). This aspect is particularly important in the luxury fashion industry, where consumers are not only looking for high-quality products, but also those that align with the brand's values and spirit (Kapferer and Michaut-Denizeau, 2014). Brands that show real commitment to sustainability forge stronger emotional connections with consumers, especially when these sustainability efforts align with their personal values (Hrebiniak, 1974; McDonald, 1981). Furthermore, the authors believe that PSV strongly influences purchase intention. When consumers believe in a brand's sustainability commitment and have an emotional tie to its values, their purchasing behavior tends to shift. This behavioral shift aligns with social exchange theory, which suggests that relationships develop and endure through reciprocal exchanges and the perceived benefits gained from those interactions (Blau, 1964). Thus, we propose the following hypotheses:

**H6b**. Perceived Sustainability Value Positively Impacts Relationship Commitment

**H6c**. Perceived Sustainability Value Positively Impacts Purchase Intention

### 2.2.7. Relationship commitment

Consumer–brand relationships are increasingly recognized as a pivotal force shaping buying behavior in sustainable and luxury markets (Morgan and Hunt, 1994; Negash and Akhbar, 2024). Drawing from Moorman et al. (1992) and Morgan and Hunt (1994), we define relationship commitment as the enduring determination of business partners to maintain a valued bond. While some scholars treat relationship commitment as a multidimensional concept (Brown, 1996; Wong and Sohal, 2002), in this study we adopt a unidimensional perspective, linking it to trust, loyalty, and satisfaction (Morgan and Hunt, 1994). When consumers perceive that a brand consistently delivers on promises—such as using blockchain to ensure product authenticity or reduce environmental impacts—they develop a deeper emotional and psychological attachment that transcends mere transactional exchanges (Treiblmaier and Garaus, 2023).

From the standpoint of relationship marketing, commitment indicates that consumers view the brand relationship as irreplaceable or uniquely beneficial (Albert and Merunka, 2013). This sense of importance arises when key brand attributes (e.g., transparency, traceability) align with consumers' personal values—particularly in markets driven

by ethical, environmental, or social concerns (Busalim et al., 2022). Because sustainable fashion choices often entail higher perceived risks (e.g., premium pricing, uncertainty about genuine eco-credentials), a brand's capacity to signal trustworthiness and authenticity through blockchain-based verification elevates consumers' sense of security and mutual dependence (Shin and Hwang, 2020). Over time, these trust signals consolidate into a robust, committed relationship.

The direct consequence of heightened commitment is a stronger behavioral inclination to purchase (Morgan and Hunt, 1994; Negash and Akhbar, 2024). Committed consumers exhibit more stable purchase patterns, prioritize the brand's offerings over competitors, and become less susceptible to short-term promotions or alternative claims of sustainability. In essence, they perceive the ongoing relationship as intrinsically valuable—an outlook that fortifies their resolve to select the brand's products (especially if those products carry meaningful sustainability or ethical assurances). Therefore, it is hypothesized that:

**H7**. Relationship commitment positively affects purchase intention.

## 3. Research methodology

### 3.1. Data collection

To ensure the validity and clarity of the survey, a pretest involving 60 respondents was initially carried out. The questionnaire was adapted from prior studies. All items were contextually reworded to reflect the second-hand luxury fashion setting while preserving their original theoretical meaning. The pilot test was conducted to assess item clarity, wording, and logical flow, leading to minor refinements before final deployment. Based on the feedback collected, the measurement scales were refined and subsequently localized into both Vietnamese and French by applying a forward and backward translation procedure. In this process, several items were carefully rephrased to enhance clarity and guarantee that the survey content aligned with the cultural context of both target groups. The survey instrument was finalized based on the pretest results and expert feedback, while SMARTPLS 4 software was later employed to conduct an analysis after data collection to validate the measurement model.

The data were gathered via an online survey, using a non-probability purposive sampling method. The use of purposive sampling limits population-level generalizability, but it is appropriate for examining theoretical relationships within the targeted group of technology-aware luxury consumers and aligns with the exploratory focus of the study. Google Forms was utilized to administer the survey. To target participants with relevant backgrounds, the survey link was circulated across social media communities focused on luxury fashion resale, as well as among customers and owners of online platforms specializing in second-hand luxury products who were familiar with blockchain concepts, including Facebook groups, Instagram pages, and private reseller networks in both Vietnam and France. Although this recruitment approach may have attracted a more digitally engaged subset of consumers, it enabled access to participants with the background needed to evaluate blockchain-enabled sustainability information. While this method may reduce representativeness, it enabled access to consumers with relevant experience in luxury fashion and blockchain-related concepts.

In terms of "luxury fashion" definition, luxury fashion lacks a universally agreed-upon conceptualization, as scholars argue that luxury is a subjective construct that varies based on individual perceptions (Kapferer and Michaut-Denizeau, 2014; Chandon et al., 2016; Janssen et al., 2017). Prior research emphasizes that luxury is not defined solely by price; instead, it encompasses perceived exclusivity, superior craftsmanship, symbolic value, brand prestige, and creative design (Phau and Prendergast, 2000; De Barnier et al., 2012; Kapferer and Michaut-Denizeau, 2014; Serdari, 2020). However, the absolute price consumers perceive as "luxury" differs substantially between Vietnam and France due to disparities in purchasing power and market maturity. Accordingly, this study adopts a category-based operational definition of luxury fashion that aligns with both theoretical criteria and market realities in Vietnam and France, rather than an absolute price cutoff. While this category-based approach improves cross-country comparability, it may mask within-country heterogeneity in luxury perceptions (e.g., differences in luxury involvement, brand knowledge, or willingness-to-pay), which we address as a limitation and avenue for future research. Respondents were considered "luxury fashion consumers" if they had previously purchased products belonging to either (1) globally recognized luxury brands characterized by exceptional craftsmanship, heritage, exclusivity, and premium pricing (e.g., Hermès, Chanel, Dior, Louis Vuitton), or (2) internationally acknowledged accessible luxury brands, which retain symbolic value, design differentiation, and higher price points relative to mainstream fashion (e.g., Lacoste, Longchamp, Sandro, DKNY, Puma). This approach aligns with cross-national consumer research standards by preserving conceptual and experiential equivalence across samples while avoiding socio-economic bias in participant eligibility.

To determine eligibility, respondents were required to meet two inclusion criteria:

(1) having previously purchased luxury fashion items, and
(2) being familiar with blockchain technology.

The verbatim screening questions used were:

(1) "Have you ever purchased luxury fashion items?"
(2) "Have you ever experienced blockchain-enabled technologies in authenticating information or reviews about the products?"

Participants who answered "No" to either question were automatically excluded. To ensure participants clearly understood the study, they were first given a short introduction that included a simple explanation of how blockchain technologies could be applied in the second-hand luxury fashion industry. The goal was to help them grasp the key concepts before moving into the survey itself. On average, completing the survey took around 15 to 20 min. Participants were also assured that their personal information would be kept strictly confidential.

The survey was distributed through a mix of online and offline methods in both Vietnam and France. In total, 614 responses were collected, 430 from Vietnam and 184 from France. Several procedures were implemented to ensure data quality prior to analysis. Responses with excessive missing values were removed. Screening questions were also used to confirm respondents' eligibility. After screening and cleaning the data, 465 valid responses were retained, including 282 from Vietnam and 183 from France. These valid cases made up 75.7% of the total submissions. A total of 149 responses (148 from Vietnam and 1 from France) were excluded due to detected errors or inconsistencies. The final sample size of 465 respondents exceeded the recommended minimum of 410, calculated based on the rule of ten responses per item for the 41-item questionnaire, following the guideline proposed by Hair et al. (2019a,b). This also surpassed the absolute lower limit of five responses per item. To reduce potential nonresponse bias, all participants were clearly informed about the research objectives and the confidentiality of their responses.

The demographic profile of the valid respondents is summarized in Table 2. Of the total sample, approximately 46% (213 participants) were male, while 54% (252 participants) were female. In terms of age distribution, the majority (37%) were aged between 18 and 25 years, followed by 29% who were in the 26 to 30 age bracket, and only 1% were younger than 18. Regarding occupational status, employees or officers made up the largest group at 44%, followed by students at 24%, business owners or self-employed individuals at 20%, and freelancers accounting for 11%. Income levels showed variation between the two countries. In Vietnam, nearly half of the respondents (49%) reported monthly earnings below 10 million VND, 21% earned between 10 and 20 million

**Table 2**
Respondent's demographic status (n = 465).

| Dimensions | Vietnam (n = 282) | | Dimensions | France (n = 183) | |
|---|---|---|---|---|---|
| | F | % | | F | % |
| ***Gender*** | | | | | |
| Male | 111 | 39 | Male | 102 | 56 |
| Female | 171 | 61 | Female | 81 | 44 |
| ***Age*** | | | | | |
| Under 18 years old | 3 | 1 | Under 18 years old | 0 | 0 |
| 18 to 25 years old | 136 | 48 | 18 to 25 years old | 34 | 19 |
| 26 to 30 years old | 64 | 23 | 26 to 30 years old | 73 | 40 |
| 31 to 35 years old | 27 | 10 | 31 to 35 years old | 54 | 30 |
| 36 to 45 years old | 38 | 13 | 36 to 45 years old | 19 | 10 |
| Over 45 years old | 14 | 5 | Over 45 years old | 3 | 2 |
| ***Occupation*** | | | | | |
| Student | 109 | 39 | Student | 4 | 2 |
| Freelancer | 31 | 11 | Freelancer | 19 | 10 |
| Business owner/Self-employed | 25 | 9 | Business owner/Self-employed | 68 | 37 |
| Employee/Officer | 107 | 38 | Employee/Officer | 80 | 44 |
| Technician/IT specialist/ Technology professional | 10 | 4 | Technician/IT specialist/ Technology professional | 10 | 5 |
| Homemaker | 0 | 0 | Homemaker | 2 | 1 |
| ***Income*** | | | | | |
| Under 10 million VND | 137 | 49 | Under $500 USD | 4 | 2 |
| From 10 to 20 million VND | 58 | 21 | $500 - $1000 USD | 10 | 5 |
| From 21 to 30 million VND | 40 | 14 | $1001 - $2000 USD | 69 | 38 |
| From 31 to 40 million VND | 16 | 6 | $2001 - $5000 USD | 80 | 44 |
| From 41 to 50 million VND | 5 | 2 | $5001 - $10,000 USD | 14 | 8 |
| From 51 to 60 million VND | 6 | 2 | Over $10,000 USD | 6 | 3 |
| Over 60 million VND | 20 | 7 | | | |

VND, and 14% reported earnings between 21 and 30 million VND, while 7% exceeded 60 million VND per month. In France, 44% of participants earned between $2001 and $5000 monthly, 38% fell within the $1001 to $2000 range, and only 2% earned less than $500.

As data were gathered using an online voluntary survey, self-selection bias could not be fully eliminated. Participants with greater interest in luxury fashion or higher familiarity with blockchain technologies may have been more motivated to participate. This may limit the generalizability of findings to the broader population of luxury consumers. However, given that the study required respondents with sufficient domain knowledge to evaluate blockchain-enabled sustainability information, such sampling remains appropriate for exploratory cross-national research. Participants were informed of the study's purpose, anonymity, and voluntary nature, and proceeding to the next screen constituted informed consent. No personal identifiers were collected, and data were stored securely in accordance with standard data privacy protocols. Anonymized data may be made available upon reasonable request, and the SmartPLS procedures are fully replicable as described.

### 3.2. Measurement

A five-point Likert scale was utilized, with response options ranging from “strongly disagree” (1) to “strongly agree” (5). This format was chosen for its simplicity and user-friendly nature, requiring less time and cognitive effort from participants compared to scales with more response categories. We decided to use a five-point scale for its ease of use on mobile devices, helping to minimize respondent fatigue while still providing sufficient detail to capture meaningful differences in responses. To ensure the quality and reliability of the measurement model, the study incorporated both established scales and newly developed items. In particular, the construct for BES-I) was adapted from Hina et al. (2024a,b) and Sadiq et al. (2023). Similarly, the construct for BES-CIT was based on measurement scales developed by Hina et al. (2024a,b) and Sadiq et al. (2023). For the BES-CCIC construct, items were also adapted from the same two studies. The measurement for BE-EWOM was based on the scale developed by Le et al. (2024). Regarding constructs related to social and perceptual dimensions, CT was adapted from Shin (2010). PI was measured using established scales from Han (2019) and Yang et al. (2016). RC followed the original conceptualization by Morgan and Hunt (1994), with adaptations by Munnukka and Järvi (2015). Finally, the construct for PSV was derived from Hamari et al. (2016). Full item wordings and adaptation details were included in Appendix.

### 3.3. Data analysis

To evaluate the proposed theoretical model and test the research hypotheses (see Fig. 1), Structural Equation Modeling (SEM) was conducted using SMARTPLS 4.0. Specifically, this study applied the Partial Least Squares Structural Equation Modeling (PLS-SEM) approach, supported by the SMARTPLS 4 software (Ringle et al., 2022). Given its ability to capture complex associations while supporting predictive analysis, PLS-SEM is particularly advantageous for studies aiming to achieve both explanatory insight and forecasting capability (Lim et al., 2023). One of the key advantages of PLS-SEM is that it does not rely on the assumption of multivariate normality, making it particularly suitable for analyzing complex models with multiple constructs and structural paths, especially when working with smaller or non-normally distributed samples (Hair et al., 2019a,b). PLS-SEM estimates linear relationships between latent constructs. Prior to model estimation, the data were examined for extreme nonlinearity, and no strong deviations from linear patterns were observed. Therefore, a linear modeling approach was considered appropriate for addressing the research objectives of this study. In addition, PLS-SEM is well-equipped to handle advanced model structures, including those that involve moderating effects and higher-order constructs composed of both reflective and formative indicators (Becker et al., 2022). Another strength of PLS-SEM lies in its ability to analyze multiple independent variables at once, as well as estimate path models based on composite constructs (Hair et al., 2019a,b). Furthermore, as this study involved a comparison between two national samples, Multi-Group Analysis (MGA) was performed to rigorously examine potential differences in the structural relationships across countries. The analysis was conducted in SMARTPLS following a two-step procedure: first, measurement invariance was assessed to ensure that the constructs were comparable between the Vietnamese and French samples; second, the path coefficients were compared across groups. This approach enabled the study to determine whether the hypothesized relationships varied significantly between the two national contexts (see Fig. 2).

The analytical procedure comprised two main stages. In the first stage, the measurement model was evaluated to confirm reliability and validity, including internal consistency, convergent validity, and discriminant validity. In the second stage, the structural model was assessed to test the hypothesized relationships and evaluate the statistical significance of the path coefficients. For the MGA, the Measurement Invariance of Composite Models (MICOM) procedure was applied to verify configural and compositional invariance, in accordance with Henseler et al. (2016). After establishing partial or full measurement invariance, cross-group differences in path coefficients were examined using the PLS-MGA procedure, allowing for the identification of statistically significant variations in the model across the Vietnamese and French samples. The significance of the structural paths was evaluated using bootstrapping with 5000 resamples, two-tailed testing at a 95% confidence level. Standard criteria were applied, whereby t-values

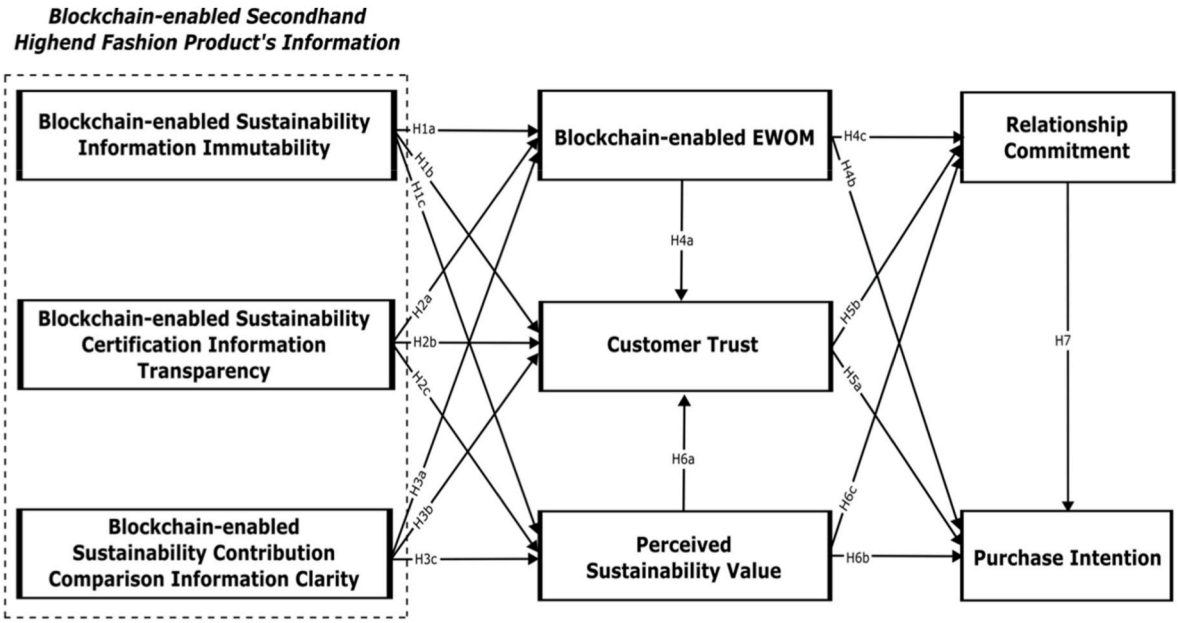


**Fig. 1.** Research model. (Authors' construction).

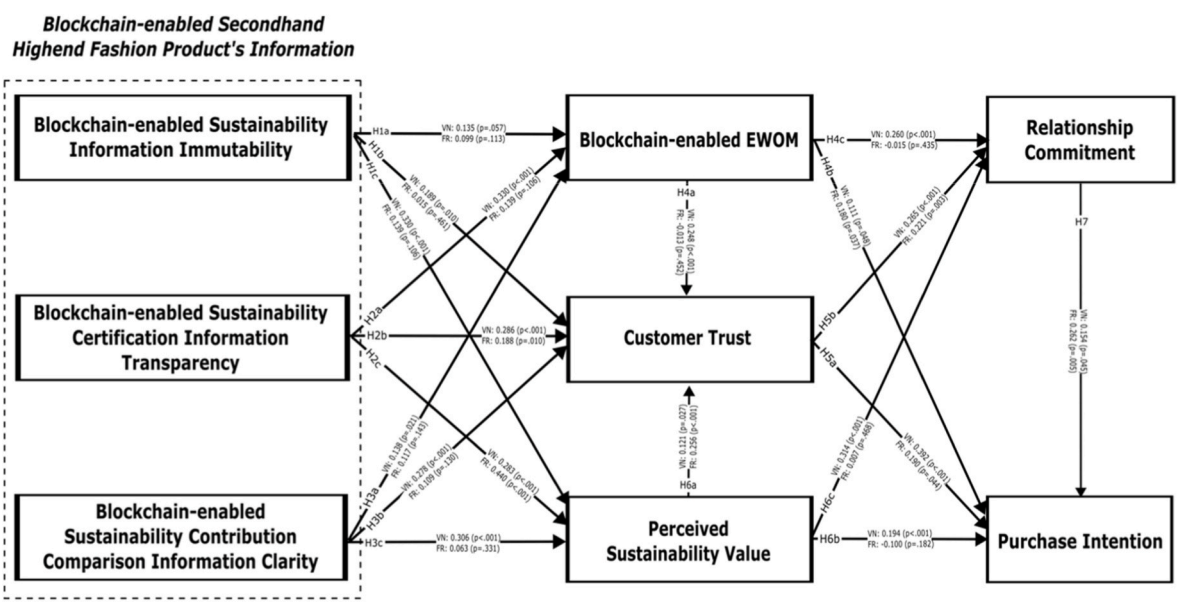


**Fig. 2.** Hypothesis testing results.

above 1.96 ($p < 0.05$) indicate statistical significance. For transparency, the structural results report the original estimates, bootstrap means, standard deviations, t-statistics, p-values, and 95% BCa confidence intervals.

## 4. Results

### 4.1. Common method bias (CMB)

Common Method Bias (CMB) may occur when data for both independent and dependent variables are collected from the same source, potentially inflating or deflating the observed relationships (MacKenzie and Podsakoff, 2012). To mitigate this issue, both procedural and statistical controls were applied in line with established recommendations (Baumgartner and Weijters, 2012; MacKenzie and Podsakoff, 2012; Podsakoff et al., 2012). Procedurally, several safeguards were incorporated during the design phase of the questionnaire. These included assuring respondents of their anonymity, emphasizing that there were no right or wrong answers, and employing neutral, non-leading language. The survey was also kept concise to minimize respondent fatigue. Furthermore, item formats were diversified, for instance, by including binary (yes/no) questions - to reduce acquiescence bias and discourage patterned responses. Statistically, multiple diagnostic techniques were employed to assess the extent of CMB. First, Harman's single-factor test was conducted, and the largest variance explained by any single factor was 36.714%, which is well below the 50% threshold, indicating that CMB is unlikely to be a major concern (MacKenzie and Podsakoff, 2012). Second, full collinearity was assessed using the Variance Inflation Factor (VIF) method. All constructs exhibited VIF values between 1.273 and 2.102, remaining well below the conservative cut-off value of 3.33 indicating that multicollinearity and CMB are unlikely to be problematic, as suggested by Kock and Lynn

(2012). Third, a marker variable technique was employed to statistically confirm that common method variance did not significantly distort the results. Correlations between the marker variable and the main constructs were found to be low and non-significant, supporting the conclusion that CMB did not substantially bias the results. The combination of robust procedural design and multiple statistical diagnostics provides strong evidence that common method bias does not pose a major threat to the validity of this study's findings.

### 4.2. Measurement model

Although several items exhibited outer loadings between 0.58 and 0.69 (BES-CCIC3 = 0.609; BES-CCIC5 = 0.581; PS2 = 0.686), these indicators were retained for both theoretical and empirical reasons. Each item represents a conceptually meaningful aspect of its construct, and removing them would have reduced content validity and weakened construct coverage. Consistent with Hair et al. (2022), items with loadings between 0.40 and 0.70 may be retained when their exclusion does not yield substantial improvements in AVE or composite reliability. In our analysis, dropping these lower-loading items increased AVE or CR only marginally and did not meaningfully enhance reliability or convergent validity. Therefore, retaining these items preserves both the psychometric adequacy and the conceptual integrity of the constructs. Regarding the clarity construct (BES-CCIC), although its loadings ranged from 0.58 to 0.79, exploratory checks showed no evidence of multidimensionality or problematic cross-loadings that would justify splitting the construct. Treated as a unidimensional reflective construct, BES-CCIC demonstrated adequate reliability (CR = 0.859) and convergent validity (AVE = 0.507), and dividing it into subdimensions did not improve AVE, CR, or model fit. Regarding measurement consistency, the current study reports CR coefficients above 0.8. These values surpass the minimum acceptable threshold of 0.6 recommended by Chin (1998), as well as the stricter 0.7 standard from Henseler and Sarstedt (2013) and Bagozzi and Yi (1988). Further, a measurement scale is typically regarded as reliable when Cronbach's Alpha coefficients fall between 0.7 and 0.8. Particularly, Cronbach's Alpha scores recorded here range from 0.784 to 0.826, indicating that there is a high degree of measurement consistency across variables. Additionally, Rho_A values exceed the 0.7 benchmark (Hair et al., 2022), reinforcing internal coherence within constructs. Convergent validity was also established, as all constructs reported AVE values exceeding the recommended 0.50 threshold (Hair et al., 2022), confirming that the latent variables adequately capture the variance explained by their respective indicators (Table 3).

Furthermore, we have used several approaches to test the discriminant validity (Table 4) in order to reflect how well a construct is distinct from others within the model. Traditionally, Fornell and Larcker (1981) suggested evaluating discriminant validity using the square root of AVE. However, due to the recognized limitations of this method listed by Henseler et al. (2015), more advanced alternatives have gained wide acceptance and among them, the Heterotrait-Monotrait Ratio (HTMT), also introduced by Henseler et al. (2015), has emerged as a more reliable measure. HTMT assesses discriminant validity by comparing correlations across indicators of different constructs relative to correlations among indicators within the same construct; values closer to 1 indicate potential discriminant validity problems (Henseler et al., 2015). Additionally, Clark and Watson (1995) recommended a stricter threshold of 0.85, a criterion whose fulfillment is crucial for enhancing measurement reliability and preserves the uniqueness of each variable's insights. Importantly, the results of this study confirm that all HTMT values remain below 0.85, with HTMT values ranging from 0.440 to 0.820 and the highest value observed between CT and PI (HTMT = 0.820), thereby validating that the measured variables are not only distinct from one another but also meaningful within the given research context.

As shown in Table 5, cross-country comparisons were assessed using Cohen's d, which quantifies the magnitude of mean differences between independent groups. Following Cohen's (1988) interpretive benchmarks (0.20 = small; 0.50 = medium; 0.80 = large), most constructs exhibited small effect sizes ($|d| < 0.50$), indicating modest cross-country variation. Only BE-EWOM reached a medium effect size (d = 0.52), demonstrating a more notable difference in blockchain-enabled word-of-mouth evaluations between the two countries. Overall, these results suggest broadly comparable perception patterns across Vietnam and France, with BE-EWOM as the sole construct showing a medium level of cross-country divergence.

**Table 3**
Reliability and Validity of the study.

| Constructs | OL | CA | CR | rho_A | AVE |
|---|---|---|---|---|---|
| BE-EWOM1 | 0.849 | 0.808 | 0.874 | 0.813 | 0.635 |
| BE-EWOM2 | 0.802 | | | | |
| BE-EWOM3 | 0.767 | | | | |
| BE-EWOM4 | 0.767 | | | | |
| BES-CCIC1 | 0.795 | 0.801 | 0.859 | 0.809 | 0.507 |
| BES-CCIC2 | 0.786 | | | | |
| BES-CCIC3 | 0.609 | | | | |
| BES-CCIC4 | 0.754 | | | | |
| BES-CCIC5 | 0.581 | | | | |
| BES-CCIC6 | 0.716 | | | | |
| BES-CIT1 | 0.832 | 0.826 | 0.885 | 0.831 | 0.658 |
| BES-CIT2 | 0.854 | | | | |
| BES-CIT3 | 0.783 | | | | |
| BES-CIT4 | 0.773 | | | | |
| BES-II1 | 0.876 | 0.805 | 0.884 | 0.823 | 0.719 |
| BES-II2 | 0.876 | | | | |
| BES-II3 | 0.787 | | | | |
| CT1 | 0.868 | 0.784 | 0.874 | 0.79 | 0.699 |
| CT2 | 0.836 | | | | |
| CT3 | 0.802 | | | | |
| PI1 | 0.791 | 0.814 | 0.87 | 0.819 | 0.573 |
| PI2 | 0.781 | | | | |
| PI3 | 0.764 | | | | |
| PI4 | 0.727 | | | | |
| PI5 | 0.72 | | | | |
| PSV1 | 0.817 | 0.815 | 0.872 | 0.817 | 0.577 |
| PSV2 | 0.686 | | | | |
| PSV3 | 0.769 | | | | |
| PSV4 | 0.778 | | | | |
| PSV5 | 0.741 | | | | |
| RC1 | 0.795 | 0.789 | 0.863 | 0.79 | 0.612 |
| RC2 | 0.778 | | | | |
| RC3 | 0.756 | | | | |
| RC4 | 0.801 | | | | |

Note: OL: Outer loadings; AVE: Average variance extracted; CA: Cronbach's alpha; CR: Composite reliability; BE-EWOM: Blockchain-enable EWOM; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BES-CIT: Blockchain-enabled sustainability certification information transparency; BES-II:Blockchain-enabled sustainability information immutability; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

### 4.3. Structural model

The results of the Variance Inflation Factor (VIF) assessment revealed scores ranging from 1.273 to 2.102. According to Hair et al. (2019), VIF scores below 3 suggest that multicollinearity is not an issue. Hence, multicollinearity can be ruled out in this analysis. Regarding the coefficient of determination (presented in Table 6), values ranged between 0.429 and 0.556, indicating moderate-to-substantial explanatory power. Assessment of overall model adequacy showed acceptable consistency with observed data. The predictive relevance of the proposed model was assessed using the Stone–Geisser's $Q^2$ statistic (Stone, 1974; Geisser, 1975), which was calculated through the blindfolding procedure (Tenenhaus et al., 2005). In line with the Stone–Geisser criterion, a model demonstrates predictive capability when it can adequately predict the indicators of endogenous latent constructs. This approach integrates cross-validation and model estimation into a single procedure, emphasizing the importance of predicting observed or potential values

**Table 4**
Discriminant validity of the study.

| Heterotrait-monotrait ratio (HTMT) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | BE-EWOM | BES-CCIC | BES-CIT | BES-II | CT | PI | PSV | RC |
| **BE-EWOM** | | | | | | | | |
| **BES-CCIC** | 0.756 | | | | | | | |
| **BES-CIT** | 0.562 | 0.564 | | | | | | |
| **BES-II** | 0.491 | 0.440 | 0.772 | | | | | |
| **CT** | 0.655 | 0.743 | 0.762 | 0.695 | | | | |
| **PI** | 0.598 | 0.682 | 0.661 | 0.560 | 0.820 | | | |
| **PSV** | 0.669 | 0.727 | 0.623 | 0.550 | 0.717 | 0.644 | | |
| **RC** | 0.658 | 0.661 | 0.651 | 0.653 | 0.760 | 0.748 | 0.685 | |
| Fornell-Larcker | | | | | | | | |
| | BE-EWOM | BES-CCIC | BES-CIT | BES-II | CT | PI | PSV | RC |
| **BE-EWOM** | 0.797 | | | | | | | |
| **BES-CCIC** | 0.614 | 0.712 | | | | | | |
| **BES-CIT** | 0.464 | 0.457 | 0.811 | | | | | |
| **BES-II** | 0.404 | 0.353 | 0.636 | 0.848 | | | | |
| **CT** | 0.527 | 0.588 | 0.615 | 0.561 | 0.836 | | | |
| **PI** | 0.494 | 0.552 | 0.543 | 0.457 | 0.659 | 0.757 | | |
| **PSV** | 0.544 | 0.589 | 0.517 | 0.456 | 0.581 | 0.532 | 0.759 | |
| **RC** | 0.532 | 0.522 | 0.526 | 0.522 | 0.599 | 0.598 | 0.552 | 0.783 |

Note: BE-EWOM: Blockchain-enable EWOM; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BES-CIT: Blockchain-enabled sustainability certification information transparency; BES-II:Blockchain-enabled sustainability information immutability; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

**Table 5**
Cross-country descriptive statistics and effect sizes (Cohen's d).

| Construct | Vietnam | | France | | Cohen's d |
|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | |
| BES-II | 3.706 | 0.815 | 4.015 | 0.586 | −0.44 |
| BES-CIT | 3.858 | 0.800 | 3.950 | 0.530 | −0.13 |
| BES-CCIC | 3.771 | 0.704 | 3.536 | 0.622 | 0.35 |
| BE-EWOM | 3.707 | 0.764 | 3.337 | 0.641 | 0.52 |
| CT | 3.712 | 0.830 | 3.787 | 0.481 | −0.11 |
| PI | 3.799 | 0.745 | 3.909 | 0.484 | −0.18 |
| PSV | 3.926 | 0.753 | 3.778 | 0.530 | 0.23 |
| RC | 3.731 | 0.696 | 3.814 | 0.528 | −0.13 |

Note: Interpretation of effect sizes follows Cohen (1988). Negative values indicate higher scores for French respondents, whereas positive values indicate higher scores for Vietnamese respondents.

**Table 6**
R Square and $Q^2$_predict values.

| | R-square | R-square adjusted | $Q^2$_Predict |
|---|---|---|---|
| **BE-EWOM** | 0.429 | 0.426 | 0.413 |
| **CT** | 0.556 | 0.551 | 0.526 |
| **PI** | 0.516 | 0.512 | 0.416 |
| **PSV** | 0.441 | 0.438 | 0.401 |
| **RC** | 0.454 | 0.450 | 0.402 |

Note: BE-EWOM: Blockchain-enable EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

rather than merely estimating latent construct parameters, which may be artificial in nature (Chin, 1998; Tenenhaus et al., 2005). As all $Q^2$ values reported in Table 6 are positive, the PLS-SEM model exhibits lower prediction errors compared to a naïve benchmark based on mean values. These results confirm that the proposed model demonstrates satisfactory predictive relevance and robust out-of-sample predictive performance. The SRMR values obtained were 0.068 (for the full measurement configuration) and 0.072 (for the structured theoretical configuration), both comfortably under the recommended limit of 0.08, demonstrating satisfactory model adequacy. Additionally, the Chi-square statistics for the complete and structured configurations were 1827.590 and 1852.324, respectively. Furthermore, the obtained d_ULS and d_G metrics fell within acceptable limits, further supporting the strength of the model, detailed clearly in Table 7.

Table 8 presents the findings from the correlation assessment among the latent constructs. The results reveal significant differences between Vietnam and France in the way blockchain influences consumer trust and behavior. In Vietnam, the variables demonstrate significant interrelationships, notably the strong association between BE-EWOM and both BES-CCIC (0.747) and RC (0.720), suggesting that BE-EWOM substantially enhances the strength of the customer-brand connection. Additionally, CT has the highest correlation with PI (0.759), emphasizing that CT is a critical factor in driving PI. Furthermore, PSV exhibits a strong association with RC (0.711), suggesting that sustainability awareness contributes meaningfully to strengthening customers' dedication to staying engaged with the brand.

In contrast, in France, the correlations between variables are generally weaker. BE-EWOM has a weaker effect on CT (0.212), implying that blockchain does not significantly affect CT in this context. Notably, BES-II shows a negative correlation with BES-CCIC (−0.273) and PSV (−0.111), reflecting French consumers' skepticism about the immutability of sustainable information. However, PSV shows the highest correlation with BES-CCIC (0.442), indicating that transparency in comparing sustainable contributions is a key factor for French consumers.

Overall, blockchain has a stronger impact on trust and purchase intention in Vietnam compared to France. This disparity may stem from cultural factors, consumer behavior, and varying levels of technology acceptance in each country. Therefore, the strategy for applying blockchain in sustainable commerce should be tailored to the characteristics of each market to achieve optimal effectiveness.

The hypothesis testing was conducted using *t*-test analysis and path coefficient p-values to assess the significance of the associations between

**Table 7**
Model_Fit—fit summary.

| | Saturated model | Estimated model |
|---|---|---|
| **SRMR** | 0.068 | 0.072 |
| **d_ULS** | 2.719 | 3.061 |
| **d_G** | 0.652 | 0.67 |
| **Chi-square** | 1827.59 | 1852.324 |
| **NFI** | 0.770 | 0.766 |

**Table 8**
Latent variable correlations.

| Vietnam | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Constructs | BE-EWOM | BES-CCIC | BES-CIT | BES-II | CT | PI | PSV | RC |
| BE-EWOM | 1 | 0.747 | 0.647 | 0.675 | 0.685 | 0.698 | 0.666 | 0.72 |
| BES-CCIC | 0.747 | 1 | 0.68 | 0.672 | 0.732 | 0.73 | 0.666 | 0.724 |
| BES-CIT | 0.647 | 0.68 | 1 | 0.719 | 0.723 | 0.641 | 0.686 | 0.623 |
| BES-II | 0.675 | 0.672 | 0.719 | 1 | 0.676 | 0.583 | 0.671 | 0.673 |
| CT | 0.685 | 0.732 | 0.723 | 0.676 | 1 | 0.759 | 0.667 | 0.701 |
| PI | 0.698 | 0.73 | 0.641 | 0.583 | 0.759 | 1 | 0.691 | 0.703 |
| PSV | 0.666 | 0.666 | 0.686 | 0.671 | 0.667 | 0.691 | 1 | 0.711 |
| RC | 0.72 | 0.724 | 0.623 | 0.673 | 0.701 | 0.703 | 0.711 | 1 |
| France | | | | | | | | |
| Constructs | BE-EWOM | BES-CCIC | BES-CIT | BES-II | CT | PI | PSV | RC |
| BE-EWOM | 1 | 0.308 | 0.105 | −0.053 | 0.212 | 0.084 | 0.209 | 0.188 |
| BES-CCIC | 0.308 | 1 | −0.026 | −0.273 | 0.28 | 0.177 | 0.442 | 0.13 |
| BES-CIT | 0.105 | −0.026 | 1 | 0.373 | 0.193 | 0.175 | 0.046 | 0.22 |
| BES-II | −0.053 | −0.273 | 0.373 | 1 | 0.135 | 0.007 | −0.111 | 0.074 |
| CT | 0.212 | 0.28 | 0.193 | 0.135 | 1 | 0.225 | 0.345 | 0.253 |
| PI | 0.084 | 0.177 | 0.175 | 0.007 | 0.225 | 1 | −0.002 | 0.302 |
| PSV | 0.209 | 0.442 | 0.046 | −0.111 | 0.345 | −0.002 | 1 | 0.112 |
| RC | 0.188 | 0.13 | 0.22 | 0.074 | 0.253 | 0.302 | 0.112 | 1 |

Note: BES-II:Blockchain-enabled sustainability information immutability; BES-CIT:Blockchain-enabled sustainability certification information transparency; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BE-EWOM: Blockchain-enable EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

variables (presented in Table 9). To ensure robustness, a bootstrapping procedure with 5000 samples was employed using SMARTPLS 4. Hypotheses were considered validated when the *t*-test values exceeded 2.3 and the corresponding significance levels fell under the 0.05 threshold (Rasoolimanesh et al., 2016).

Considering Vietnamese consumer groups, most hypotheses are supported, with relationships showing p-values below 0.05 and β coefficients indicating a positive influence among the factors in the model. Specifically, hypotheses H1a ($\beta = 0.248$, $p < 0.05$), H1b ($\beta = 0.111$, $p < 0.05$), and H1c ($\beta = 0.260$, $p < 0.05$) all have p-values less than 0.05, demonstrating that BES-II exerts a considerable influence on BE-EWOM, CT, and PSV. Similarly, hypotheses regarding BES-CIT, such as H2a ($\beta = 0.138$, $p < 0.05$), H2b ($\beta = 0.278$, $p < 0.05$), and H2c ($\beta = 0.306$, $p < 0.05$), are all supported, indicating that transparency in sustainability certification information via blockchain significantly affects BE-EWOM, CT, and PSV among Vietnamese consumers. For BES-CCIC, hypotheses H3a ($\beta = 0.486$, $p < 0.05$), H3b ($\beta = 0.286$, $p < 0.05$), and H3c ($\beta = 0.283$, $p < 0.05$) are also confirmed, showing that transparency in comparing sustainability contribution information via blockchain makes a notable contribution to developing BE-EWOM, CT, and PSV. However, hypothesis H4a concerning the influence of EWOM supported by Blockchain technology on CT is not supported ($\beta = 0.135$, $p > 0.05$), implying that this form of EWOM does not exert a notable effect on the level of trust among Vietnamese consumers.

For the group of French consumers, the hypothesis testing results show clear differences compared to the Vietnamese consumer group. Specifically, among the hypotheses related to BES-II, only H1b ($\beta = 0.180$, $p < 0.05$) is supported, indicating that sustainability information through blockchain has a significant positive influence on customer trust. In contrast, H1a ($\beta = -0.013$, $p > 0.05$) and H1c ($\beta = -0.015$, $p > 0.05$) are not supported, suggesting no significant effect of BES-II on BE-EWOM and Perceived Sustainability. Regarding BES-CIT, none of the hypotheses are supported, as H2a ($\beta = 0.117$, $p > 0.05$), H2b ($\beta = 0.109$, $p > 0.05$), and H2c ($\beta = 0.063$, $p > 0.05$) all have p-values above 0.05, indicating that certification information transparency via blockchain does not significantly affect BE-EWOM, CT, or PSV among French consumers.

For BES-CCIC, hypotheses H3a ($\beta = 0.108$, $p = 0.000$), H3b ($\beta = 0.188$, $p < 0.05$), and H3c ($\beta = 0.440$, $p < 0.05$) are all supported, indicating that transparency in comparing sustainability contribution information via blockchain significantly influences EWOM, CT, and PSV among French consumers. By contrast, hypothesis H4a concerning the impact of BE-EWOM on CT ($\beta = 0.099$, $p > 0.05$) is not supported, as the p-value exceeds the 0.05 threshold, suggesting that the association between BE-EWOM and CT in the French market is not statistically significant.

The hypothesis testing results between Vietnamese and French consumer groups highlight distinct variations in how blockchain-related elements influence sustainable consumption patterns. For Vietnamese consumers, most hypotheses are supported, with positive and statistically significant relationships. Meanwhile, for French consumers, while hypotheses related to the transparency and security of sustainability information through blockchain are accepted, some hypotheses, such as the impact of BE-EWOM on CT and PSV, are not confirmed. This indicates that cultural and market factors may differently influence sustainable consumer behavior in each country.

Furthermore, we assessed out-of-sample predictive power using PLSpredict with 10-fold cross-validation (Shmueli et al., 2019). As reported in Table 10, all indicator-level $Q^2$_predict values are positive and above 0.15, so the PLS-SEM model outperforms the naïve mean benchmark for every endogenous indicator. Comparing the RMSE values with those of the linear benchmark model (LM) shows that CT and RC are strongly predicted: all of their indicators (CT1–CT3, RC1–RC4) have lower RMSE under PLS-SEM, which indicates high predictive power. For PI, three of five indicators (PI1–PI3) are better predicted by PLS-SEM than by LM, while PI4–PI5 favour LM, consistent with medium predictive power. In contrast, BE-EWOM (2 of 4 indicators) and PSV (1 of 5 indicators) only show weak incremental gains over LM, suggesting that prediction for these mediators is limited and could be improved by additional predictors or refined items (e.g., PI4–PI5, PS2–PS3).

At the latent-variable level, the CVPAT results in Table 11 show that the PLS-SEM model yields substantially lower prediction loss than the indicator-average (IA) benchmark for all endogenous constructs (all loss differences negative and significant at $p < 0.001$), confirming that the specified latent structure improves prediction over a simple aggregation of indicators. Together with the PLSpredict results, this implies that while the model clearly dominates the IA baseline, its predictive advantage over a conventional linear benchmark is concentrated in the key relationship constructs CT and RC, is moderate for PI, and remains weak for BE-EWOM and PSV.

**Table 9**
Hypothesis testing path coefficients - mean, STDEV, T-values, p-value.

| France | Original sample (O) | Sample mean (M) | Standard deviation (STDEV) | T statistics (\|O/STDEV\|) | P values | Results |
|---|---|---|---|---|---|---|
| BES-II → BE-EWOM = H1a | 0.099 | 0.096 | 0.082 | 1.209 | 0.113 | Rejected |
| BES-II → CT = H1b | 0.015 | 0.024 | 0.155 | 0.099 | 0.461 | Accepted |
| BES-II → PSV = H1c | 0.139 | 0.135 | 0.111 | 1.251 | 0.106 | Rejected |
| BES-CIT → BE-EWOM = H2a | 0.308 | 0.320 | 0.069 | 4.454 | 0.000 | Rejected |
| BES-CIT → CT = H2b | 0.188 | 0.186 | 0.081 | 2.335 | 0.010 | Rejected |
| BES-CIT → PSV = H2c | 0.440 | 0.443 | 0.063 | 6.966 | 0.000 | Rejected |
| BES-CCIC → BE-EWOM = H3a | 0.117 | 0.116 | 0.110 | 1.068 | 0.143 | Accepted |
| BES-CCIC → CT = H3b | 0.109 | 0.123 | 0.096 | 1.126 | 0.130 | Accepted |
| BES-CCIC → PSV = H3c | 0.063 | 0.041 | 0.143 | 0.438 | 0.331 | Accepted |
| BE-EWOM → CT = H4a | −0.013 | −0.014 | 0.107 | 0.122 | 0.452 | Rejected |
| BE-EWOM → PI = H4b | 0.180 | 0.175 | 0.100 | 1.791 | 0.037 | Rejected |
| BE-EWOM → RC = H4c | −0.015 | −0.020 | 0.090 | 0.165 | 0.435 | Rejected |
| CT → PI = H5a | 0.190 | 0.183 | 0.111 | 1.71 | 0.044 | Accepted |
| CT → RC = H5b | 0.221 | 0.232 | 0.082 | 2.703 | 0.003 | Accepted |
| PSV → CT = H6a | 0.256 | 0.260 | 0.077 | 3.341 | 0.000 | Accepted |
| PSV → PI = H6b | −0.100 | −0.092 | 0.110 | 0.910 | 0.182 | Rejected |
| PSV → RC = H6c | 0.007 | 0.012 | 0.085 | 0.081 | 0.468 | Rejected |
| RC → PI = H7 | 0.262 | 0.264 | 0.101 | 2.600 | 0.005 | Accepted |
| **Vietnam** | | | | | | |
| | Original sample (O) | Sample mean (M) | Standard deviation (STDEV) | T statistics (\|O/STDEV\|) | P values | Results |
| BES-II → BE-EWOM = H1a | 0.135 | 0.142 | 0.086 | 1.577 | 0.057 | Accepted |
| BES-II → CT = H1b | 0.189 | 0.197 | 0.080 | 2.345 | 0.010 | Accepted |
| BES-II → PSV = H1c | 0.330 | 0.323 | 0.073 | 4.490 | 0.000 | Accepted |
| BES-CIT → BE-EWOM = H2a | 0.486 | 0.488 | 0.053 | 9.108 | 0.000 | Accepted |
| BES-CIT → CT = H2b | 0.286 | 0.283 | 0.073 | 3.922 | 0.000 | Accepted |
| BES-CIT → PSV = H2c | 0.283 | 0.282 | 0.058 | 4.880 | 0.000 | Accepted |
| BES-CCIC → BE-EWOM = H3a | 0.138 | 0.140 | 0.068 | 2.025 | 0.021 | Accepted |
| BES-CCIC → CT = H3b | 0.278 | 0.275 | 0.079 | 3.535 | 0.000 | Accepted |
| BES-CCIC → PSV = H3c | 0.306 | 0.305 | 0.065 | 4.681 | 0.000 | Accepted |
| BE-EWOM → CT = H4a | 0.248 | 0.246 | 0.062 | 3.988 | 0.000 | Rejected |
| BE-EWOM → PI = H4b | 0.111 | 0.112 | 0.067 | 1.660 | 0.048 | Accepted |
| BE-EWOM → RC = H4c | 0.260 | 0.261 | 0.061 | 4.298 | 0.000 | Accepted |
| CT → PI = H5a | 0.392 | 0.376 | 0.113 | 3.474 | 0.000 | Accepted |
| CT → RC = H5b | 0.265 | 0.272 | 0.076 | 3.470 | 0.000 | Accepted |
| PSV → CT = H6a | 0.121 | 0.121 | 0.063 | 1.934 | 0.027 | Accepted |
| PSV → PI = H6b | 0.194 | 0.193 | 0.058 | 3.372 | 0.000 | Accepted |
| PSV → RC = H6c | 0.314 | 0.313 | 0.057 | 5.467 | 0.000 | Accepted |
| RC → PI = H7 | 0.154 | 0.165 | 0.091 | 1.694 | 0.045 | Accepted |

Note: BES-II:Blockchain-enabled sustainability information immutability; BES-CIT:Blockchain-enabled sustainability certification information transparency; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BE-EWOM: Blockchain-enable EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

### *4.4. Measurement invariance*

Before conducting cross-country comparisons, this study assessed measurement invariance between the Vietnamese and French samples using the Measurement Invariance of Composite Models (MICOM) procedure (Henseler et al., 2016a,b). Establishing measurement invariance is a prerequisite for ensuring that constructs are conceptualized and measured equivalently across groups, thereby allowing meaningful multi-group comparisons. The MICOM procedure comprises three sequential steps: (i) assessment of configural invariance, (ii) assessment of compositional invariance, and (iii) assessment of the equality of composite means and variances. First, configural invariance was established, as identical measurement models, indicators, data treatment, and algorithm settings were applied across the two national samples, and acceptable model fit was observed for both groups. Second, compositional invariance was evaluated using a permutation-based approach with 5000 resamples and two-tailed testing. As reported in Table 12, the original correlations of the composite scores were close to 1, and the corresponding 95% confidence intervals included the value of 1 for all constructs. In addition, all permutation p-values exceeded the 0.05 threshold, indicating that compositional invariance was established across the Vietnamese and French samples. Third, the equality of composite mean values and variances was examined. The permutation-based confidence intervals for mean and variance differences included zero for all constructs, suggesting no statistically significant differences in composite means or variances between the two groups.

MICOM results confirm configural and compositional invariance and therefore establish partial measurement invariance, which is sufficient for conducting cross-group comparisons of structural path coefficients (Henseler et al., 2016a,b). Although the equality of composite means and variances is also supported, the present study focuses on differences in structural relationships rather than latent mean comparisons. Consequently, multi-group analysis (MGA) was conducted to examine whether the hypothesized relationships differ between Vietnam and France.

### *4.5. Multi-group analysis*

To examine potential cross-country differences in the structural relationships, a multi-group analysis (MGA) was conducted using a permutation-based PLS-MGA approach with 5000 resamples and two-tailed testing. This method allows for a robust comparison of path coefficients across groups while accounting for sampling variability. The MGA results are presented in Table 13.

Overall, the findings reveal that while many structural relationships are statistically similar across the two countries, several paths exhibit significant cross-country differences. In particular, the effect of blockchain-enabled electronic word-of-mouth (BE-EWOM) on

**Table 10**
PLSpredict assessment of manifest variables (original model).

| | PLS-SEM | | LM | PLS-SEM - LM |
|---|---|---|---|---|
| | $Q^2$predict | RMSE | RMSE | RMSE |
| BE-EWOM1 | 0.312 | 0.767 | 0.76 | 0.007 |
| BE-EWOM2 | 0.275 | 0.751 | 0.75 | 0.001 |
| BE-EWOM3 | 0.219 | 0.917 | 0.919 | −0.002 |
| BE-EWOM4 | 0.237 | 0.761 | 0.78 | −0.019 |
| PI1 | 0.33 | 0.713 | 0.719 | −0.006 |
| PI2 | 0.26 | 0.724 | 0.73 | −0.006 |
| PI3 | 0.242 | 0.779 | 0.788 | −0.009 |
| PI4 | 0.157 | 0.782 | 0.759 | 0.023 |
| PI5 | 0.188 | 0.791 | 0.773 | 0.018 |
| CT1 | 0.436 | 0.649 | 0.657 | −0.008 |
| CT2 | 0.368 | 0.646 | 0.651 | −0.005 |
| CT3 | 0.29 | 0.747 | 0.756 | −0.009 |
| RC1 | 0.253 | 0.722 | 0.735 | −0.013 |
| RC2 | 0.253 | 0.688 | 0.689 | −0.001 |
| RC3 | 0.235 | 0.704 | 0.708 | −0.004 |
| RC4 | 0.241 | 0.71 | 0.724 | −0.014 |
| PSV1 | 0.286 | 0.761 | 0.755 | 0.006 |
| PSV2 | 0.203 | 0.753 | 0.717 | 0.036 |
| PSV3 | 0.201 | 0.865 | 0.859 | 0.006 |
| PSV4 | 0.229 | 0.794 | 0.798 | −0.004 |
| PSV5 | 0.224 | 0.742 | 0.741 | 0.001 |

Note: BES-II:Blockchain-enabled sustainability information immutability; BES-CIT:Blockchain-enabled sustainability certification information transparency; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BE-EWOM: Blockchain-enable EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

**Table 11**
CVPAT LV Summary - PLS-SEM vs. Linear Model (LM).

| | PLS loss | IA loss | Average loss difference | t value | p value |
|---|---|---|---|---|---|
| **BE-EWOM** | 0.643 | 0.867 | −0.224 | 6.629 | 0.000 |
| **PI** | 0.575 | 0.753 | −0.177 | 5.277 | 0.000 |
| **CT** | 0.465 | 0.731 | −0.266 | 6.415 | 0.000 |
| **RC** | 0.499 | 0.661 | −0.162 | 5.449 | 0.000 |
| **PSV** | 0.615 | 0.797 | −0.182 | 5.927 | 0.000 |
| **Overall** | 0.567 | 0.764 | −0.197 | 6.657 | 0.000 |

Note: BES-II: Blockchain-enabled sustainability information immutability; BES-CIT: Blockchain-enabled sustainability certification information transparency; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BE-EWOM: Blockchain-enabled EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

relationship commitment is significantly stronger in Vietnam than in France ($\beta = 0.260$, $p = 0.032$). This result suggests that BE-EWOM plays a more central role in fostering long-term relational bonds with consumers in the Vietnamese market. Significant differences are also observed for the effect of blockchain-enabled sustainability contribution comparison information clarity (BES-CCIC) on purchase intention ($\beta = 0.283$, $p = 0.001$). The stronger effect in Vietnam indicates that clear and comparable sustainability contribution information exerts a greater influence on consumers' purchase intentions in an emerging market context, where such information may reduce uncertainty and enhance decision confidence. Similarly, the relationship between purchase intention and perceived sustainability value differs significantly between the two countries ($\beta = 0.335$, $p = 0.001$), indicating that intentions to purchase translate more strongly into sustainability value perceptions among Vietnamese consumers.

In addition, the effect of relationship commitment on perceived sustainability value is significantly stronger in Vietnam than in France ($\beta = 0.265$, $p = 0.007$), highlighting the greater importance of relational mechanisms in shaping sustainability-related evaluations in the Vietnamese context. By contrast, several structural paths do not exhibit statistically significant cross-country differences. For example, the effects of BE-EWOM on customer trust and perceived sustainability value, as well as the effects of blockchain-enabled sustainability information immutability (BES-II) and certification information transparency (BES-CIT) on downstream outcomes, are largely comparable across the two samples. These results suggest that, despite contextual differences, certain blockchain-enabled communication mechanisms operate similarly in both countries. In sum, the MGA results indicate that cross-country differences primarily emerge in relational and intention-driven mechanisms rather than in foundational trust-formation processes. This pattern underscores the context-dependent role of blockchain-enabled sustainability communication, with stronger relational and behavioral transmission effects observed in Vietnam relative to France.

## 5. Discussion

The research seeks to explore the mechanisms through which BES-II enhances EWOM, CT, and PSV, and how these drivers affect consumers' intention to purchase second-hand luxury fashion goods.

First, the results reveal notable differences in how BES-II influences BE-EWOM, CT, and PSV across Vietnamese and French consumers. For Vietnamese consumers participating in this study, BES-II significantly enhances BE-EWOM (H1a: $\beta = 0.248$, $p < 0.05$), CT (H1b: $\beta = 0.111$, $p < 0.05$), and PSV (H1c: $\beta = 0.260$, $p < 0.05$). These findings align with prior research emphasizing the role of information security in strengthening trust and sustainability perceptions (Kim and Peterson, 2017). In contrast, among French consumers participating in this study, Blockchain-enabled Sustainability Information Immutability does not meaningfully affect BE-EWOM (H1a: $\beta = -0.013$, $p > 0.05$) or PSV (H1c: $\beta = -0.015$, $p > 0.05$), although it positively influences CT (H1b: $\beta = 0.180$, $p < 0.05$). One plausible macro-level explanation is that the two markets differ in technology readiness and digital maturity (Abudu and Agbeko, 2025; Chan et al., 2023; Ungson and Soorapanth, 2022). Evidence from emerging economies shows that lower digital maturity and weaker institutional support make security-enhancing features such as blockchain immutability and traceability particularly salient as risk-reducing cues, especially in supply chains where authenticity, provenance and sustainability are contested (Tan et al., 2025; Wolf et al., 2022; Ahmed and MacCarthy, 2022). By contrast, in more digitally mature and tightly regulated contexts, consumer trust tends to be anchored in regulatory support, established labels and institutional safeguards rather than in any single underlying technology (Albayati et al., 2020; Xiao et al., 2024), so blockchain often operates as one background infrastructure among others rather than as a primary differentiator.

Second, the findings offer a more refined understanding of how BES-CIT influences BE-EWOM, CT, and PSV across the two markets. In Vietnam, greater transparency in sustainability-related information exerts significant positive effects on BE-EWOM (H2a: $\beta = 0.138$, $p < 0.05$), reinforces CT (H2b: $\beta = 0.278$, $p < 0.05$), and enhances PSV (H2c: $\beta = 0.306$, $p < 0.05$). These results are consistent with Shin and Hwang (2020), who emphasize that heightened transparency contributes to stronger customer trust and more favorable evaluations. Conversely, in the French market, BES-CIT does not exhibit any statistically significant influence on BE-EWOM (H2a: $\beta = 0.117$, $p > 0.05$), CT (H2b: $\beta = 0.109$, $p > 0.05$), or PSV (H2c: $\beta = 0.063$, $p > 0.05$). Although previous research, Ying et al. (2023) and Hina et al. (2024a,b), highlights the value of blockchain-enabled transparency in improving consumers' sustainability assessments, its limited impact in France may be attributed to consumers' stronger reliance on well-established eco-labels and fair-trade certifications. This interpretation aligns with Brécard et al. (2012), who demonstrate that French consumers tend to trust traditional sustainability indicators more than emerging digital transparency mechanisms.

Third, the analysis demonstrates that BES-CCIC exerts a significant

**Table 12**
Measurement invariance result using permutation test.

| Compositional Invariance Correlation = 1 | | | | | Equal Mean Assessment | | | Equal Variance Assessment | |
|---|---|---|---|---|---|---|---|---|---|
| Construct | Configure Invariance | C = 1 | 95% CI | Partial Measurement Invariance Established | 95% Confidence Interval | Equal Mean | 95% CI | Equal Variance | Full Measurement Invariance Established |
| **BES-II** | Yes | 0.996 | 0.983-1.000 | Yes | [-0.224; 0.238] | Yes | [-0.278; 0.247] | Yes | Yes |
| **BES-CIT** | Yes | 1 | 0.993-1.000 | Yes | [-0.247; 0.258] | Yes | [-0.308; 0.277] | Yes | Yes |
| **BES-CCIC** | Yes | 0.999 | 0.997-1.000 | Yes | [-0.232; 0.229] | Yes | [-0.297; 0.278] | Yes | Yes |
| **BE-EWOM** | Yes | 0.999 | 0.997-1.000 | Yes | [-0.213; 0.219] | Yes | [-0.249; 0.280] | Yes | Yes |
| **CT** | Yes | 0.996 | 0.999-1.000 | Yes | [-0.243; 0.279] | Yes | [-0.285; 0.202] | Yes | Yes |
| **PI** | Yes | 1 | 0.992-1.000 | Yes | [-0.293; 0.270] | Yes | [-0.303; 0.198] | Yes | Yes |
| **PS** | Yes | 1 | 0.996-1.000 | Yes | [-0.211; 0.189] | Yes | [-0.203; 0.257] | Yes | Yes |
| **RC** | Yes | 0.998 | 0.989-1.000 | Yes | [-0.223; 0.209] | Yes | [-0.233; 0.249] | Yes | Yes |

Note: BES-II:Blockchain-enabled sustainability information immutability; BES-CIT:Blockchain-enabled sustainability certification information transparency; BES-CCIC: Blockchain-enabled sustainability contribution comparison information clarity; BE-EWOM: Blockchain-enable EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

**Table 13**
Assessment of group difference Vietnam - France (MGA).

| Hypotheses | Path Coefficient Difference | p value |
|---|---|---|
| **BE-EWOM - > CT** | −0.04 | 0.697 |
| **BE-EWOM - > PSV** | −0.005 | 0.956 |
| **BE-EWOM - > RC** | 0.26 | 0.032* |
| **BES-CCIC - > BE-EWOM** | 0.174 | 0.032* |
| **BES-CCIC - > CT** | −0.119 | 0.218 |
| **BES-CCIC - > PI** | 0.283 | 0.001* |
| **BES-CIT - > BE-EWOM** | 0.046 | 0.685 |
| **BES-CIT - > CT** | 0.157 | 0.134 |
| **BES-CIT - > PI** | 0.003 | 0.984 |
| **BES-II - > BE-EWOM** | 0.246 | 0.016* |
| **BES-II - > CT** | −0.054 | 0.532 |
| **BES-II - > PI** | 0.082 | 0.532 |
| **CT - > RC** | 0.078 | 0.474 |
| **PI - > CT** | 0.231 | 0.028* |
| **PI - > PSV** | 0.335 | 0.001* |
| **PI - > RC** | 0.01 | 0.939 |
| **RC - > PSV** | 0.265 | 0.007* |

Note: *The Differences are significant in the relationships between the two countries ($P < 0.05$).
Note: BES-II:Blockchain-enabled sustainability information immutability; BES-CIT:Blockchain-enabled sustainability certification information transparency; BES-CCIC Blockchain-enabled sustainability contribution comparison information clarity; BE-EWOM: Blockchain-enable EWOM; CT: Customer Trust; PI: Purchase Intention; PSV: Perceived Sustainability Value; RC: Relationship commitment.

positive effect on BE-EWOM in both Vietnam (H3a: $\beta = 0.486$, $p < 0.05$) and France (H3a: $\beta = 0.308$, $p < 0.05$). These findings align with Sadiq et al. (2023), who argued that AI-generated sustainability comparisons can enhance consumers' evaluative responses. The present study extends this line of evidence by showing that comparison clarity enabled through blockchain technology can similarly stimulate positive electronic word-of-mouth. This result diverges from Mazzù et al. (2023) who suggested that blockchain-generated information may be perceived as less credible than information from human sources. A key contribution of this research lies in isolating the effect of comparison clarity specifically within a blockchain-enabled environment. Furthermore, BES-CCIC is a strong predictor of CT in both markets, Vietnam (H3b: $\beta = 0.286$, $p < 0.05$) and France (H3b: $\beta = 0.188$, $p < 0.05$), which is consistent with Watson et al. (2010) and Ying et al. (2023). The construct also significantly enhances PSV in Vietnam (H3c: $\beta = 0.283$, $p < 0.05$) and France (H3c: $\beta = 0.440$, $p < 0.05$), supporting prior evidence that blockchain-enabled clarity can improve consumers' sustainability evaluations (Hina et al., 2024a,b).

Fourth, the findings reveal that BE-EWOM does not exert a significant influence on CT in either Vietnam ($\beta = 0.135$, $p > 0.05$) or France ($\beta = 0.099$, $p > 0.05$), resulting in the rejection of H4a. This outcome contrasts with prior studies such as Bulut and Karabulut (2018) and Anser et al. (2020), which reported that strong EWOM can enhance customer trust. A plausible explanation is that although blockchain ensures data integrity, consumers may not yet fully recognize or internalize its credibility-enhancing properties (Jain et al., 2020). In the Vietnamese context, trust formation tends to rely more heavily on social interaction, relational ties, and interpersonal validation (Ye et al., 2019), which may reduce the relative weight of blockchain-based EWOM. In France, the absence of a significant effect may reflect consumers' reliance on institutional trust frameworks and well-established certification systems, as noted by Komdeur and Ingenbleek (2021). These findings suggest that blockchain-enabled EWOM, on its own, may be insufficient to foster trust in markets where traditional credibility cues continue to dominate.

In contrast, BE-EWOM appears to operate more as an informational cue that supports behavioural intentions and relationship outcomes, particularly in Vietnam. Our findings confirm H4b for Vietnam ($\beta = 0.189$, $p < 0.05$), indicating that BE-EWOM positively affects PI, which is consistent with Rahman et al. (2020) and Khan et al. (2023). However, in France, the effect is non-significant ($\beta = 0.015$, $p > 0.05$), suggesting that immutable EWOM does not necessarily translate into stronger purchase intentions, potentially due to cultural differences in how credibility is constructed. BE-EWOM also strengthens RC in Vietnam, where access to verified peer feedback facilitates relationship-building, aligning with Izogo et al. (2022); Bulut and Karabulut (2018) and Kushwaha et al. (2020). In France, the weaker effect may reflect skepticism toward immutable sustainability claims and a greater reliance on institutional trust. Taken together, this finding suggests that BE-EWOM functions more as an informational cue influencing behavioral intentions and relational outcomes rather than as a direct trust-building mechanism. This extends prior EWOM literature by showing that in blockchain-enabled sustainability contexts, trust formation remains anchored in institutional and relational foundations, while technologically verified peer information primarily supports decision facilitation rather than credibility formation.

Fifth, the results indicate that CT significantly enhances PI in both

markets, with a stronger effect in Vietnam ($\beta = 0.392$) than in France ($\beta = 0.190$). This is consistent with Nguyen and Nguyen (2021) and Duong et al. (2024), who demonstrated that trust reduces perceived risk and strengthens purchase motivation. Interestingly, Meira et al. (2024) suggest that a slight degree of uncertainty can also stimulate purchase intention by fostering curiosity. Ramkumar and Jin (2019), Zhu et al. (2019), and Watanabe et al. (2020) further argue that consumers sometimes rely less on certifications and more on other cues when making purchase decisions. Although trust has been explored across various contexts, its interplay with relational commitment and purchase intention remains underexamined in blockchain-enabled sustainability settings. This study is among the first to empirically demonstrate how relational commitment can enhance purchasing intention in this emerging domain, offering valuable insights into evolving consumer behavior.

Sixth, CT positively influences RC in both Vietnam (H5b: $\beta = 0.265$; $p < 0.05$) and France (H5b: $\beta = 0.221$; $p < 0.05$). This supports Lin et al. (2022) and Wistedt (2024), underscoring trust as a central factor in sustaining long-term brand relationships. However, while previous studies have examined digital trust in general, this research uniquely integrates BE-EWOM as a trust-enhancing mechanism. This study is one of the first to empirically validate the influence of blockchain-driven transparency on RC, contributing new insights to the field.

Seven, the analysis examines how PSV influences CT, RC, and PI. For Vietnamese consumers, PSV significantly affects CT (H6a: $\beta = 0.121$, $p < 0.05$), RC (H6c: $\beta = 0.314$, $p < 0.05$), and PI (H6b: $\beta = 0.194$, $p < 0.05$), consistent with Zhuang et al. (2021) and Wang et al. (2024). Among French consumers, PSV significantly predicts CT (H6a: $\beta = 0.256$, $p < 0.05$), but not PI (H6b: $\beta = -0.100$, $p > 0.05$) or RC (H6c: $\beta = 0.007$, $p > 0.05$). These findings suggest that while sustainability values enhance trust, they do not necessarily drive behavioral outcomes in developed markets, challenging assumptions of universal sustainability effects (Wang et al., 2021; Al-Issa, 2024; Balabanis et al., 2024). Overall, the results extend existing literature by highlighting market-specific differences in how sustainability perceptions translate into consumer action.

Eighth, RC positively affects PI in both Vietnam (H7: $\beta = 0.154$; $p < 0.05$) and France (H7: $\beta = 0.262$; $p < 0.05$). This finding is consistent with prior research (Krolikowski and Yuan, 2017; Cui et al., 2019; Lin et al., 2022; Wistedt, 2024), which suggests that relationship commitment fosters continued purchase behavior. While Rehman et al. (2019) noted that this relationship may depend on factors such as brand image, this study offers new insights by examining RC within blockchain-supported markets. In such contexts, the transparency afforded by blockchain enhances trust and facilitates informed purchasing decisions, reinforcing the link between commitment and intentional buying.

Finally, the multi-group analysis further enriches the discussion by revealing how the mechanisms through which blockchain-enabled sustainability communication influences consumer responses vary across national contexts. Although the MICOM results indicate that the constructs are measured equivalently across Vietnam and France, the MGA findings suggest that the strength of several structural relationships is context-dependent rather than universal. Notably, the stronger effects observed in Vietnam for several relational and intention-driven paths point to the heightened role of blockchain-enabled communication in shaping consumer behavior in an emerging market context. For instance, the significantly stronger impact of blockchain-enabled electronic word-of-mouth on relationship commitment in Vietnam suggests that consumers in this market rely more heavily on technologically mediated social signals to form durable relational bonds. This finding aligns with prior research indicating that, in markets characterized by higher uncertainty and information asymmetry, digitally enabled credibility cues and peer-based communication play a more prominent role in reducing perceived risk and fostering trust-based relationships. Similarly, the stronger influence of sustainability contribution comparison information clarity on purchase intention, as well as the stronger translation of purchase intention into perceived sustainability value in Vietnam, underscores the importance of clear, comparable, and actionable sustainability information in guiding consumer decision-making. In contexts where sustainability standards and third-party verification mechanisms are still evolving, blockchain-enabled transparency may serve as a critical substitute for institutional trust, thereby amplifying its behavioral impact. In contrast, French consumers - operating within a more mature regulatory and informational environment - may already take such transparency for granted, resulting in weaker marginal effects. By comparison, several foundational relationships, such as the effects of blockchain-enabled sustainability information immutability and certification transparency on trust-related outcomes, do not differ significantly between the two countries. This pattern suggests that certain core mechanisms of blockchain-enabled sustainability communication operate similarly across contexts, reflecting a degree of structural stability in how consumers process credibility-enhancing technological features. The MGA findings indicate that cross-country differences are less pronounced at the level of basic trust formation and more salient in downstream relational and behavioral pathways. This distinction highlights the importance of considering market maturity and contextual uncertainty when theorizing about the effects of emerging technologies in sustainability communication. Rather than assuming uniform impacts across countries, future research should further explore how institutional development, digital literacy, and cultural orientations toward technology condition the effectiveness of blockchain-enabled communication strategies.

## 6. Implications

This study examines how three key blockchain affordances (immutability, transparency, and clarity) shape customer trust, perceived sustainability value, electronic word-of-mouth, and purchase intention in the second-hand luxury fashion market in Vietnam and France. The findings reveal substantial cross-cultural differences in how consumers interpret information about blockchain-enabled sustainability. While immutability and transparency enhance trust and sustainability perceptions more strongly among Vietnamese consumers, French consumers remain more reliant on traditional certifications. Clarity emerges as a consistent predictor of sustainability-related EWOM in both countries, emphasizing the importance of easily interpretable impact data. Additionally, the study reveals culturally contingent mechanisms underlying the influence of BE-EWOM, sustainability value, and relational commitment on purchase intention. These insights advance theoretical understanding of blockchain-enabled sustainability communication while offering actionable guidance for brands and policymakers operating in culturally diverse markets.

### 6.1. Theoretical implications

First, our study offers important theoretical findings by extending Theory of Affordance through examining blockchain features (Immutability, Transparency and Clarity) as well as illustrating how these features influence consumer behavior within the second-hand luxury fashion market across two different countries, Vietnam and France. Our research conceptualizes clarity as a distinct affordance of blockchain-based sustainability communication, in addition to the well-established affordances of transparency and immutability (Shin and Hwang, 2020; Jahanbin et al., 2023). While prior studies emphasized transparency and traceability, our findings demonstrate that clarity - defined as the interpretability of complex sustainability information - plays a critical role in enabling consumers to comprehend and act upon blockchain-enabled data (Sharma et al., 2020; Panghal et al., 2023; Isabel et al., 2024). Furthermore, by showing that affordances interact with cultural differences to yield divergent consumer interpretations, our study refines affordance theory to highlight the cultural contingency

of technological affordances (Volkoff and Strong, 2017).

Second, the results further offer novel insights into the relational dynamics between blockchain features and key consumer-related constructs, including electronic word-of-mouth (EWOM), customer trust (CT), and perceived sustainability value (PSV). Specifically, immutability was found to enhance trust among Vietnamese consumers but exhibited a limited effect in the French context. Similarly, transparency positively influenced CT and PSV in Vietnam, but showed no significant effect in France, where institutional trust appears to rely more on traditional certifications. These differences reflect broader cultural and institutional trust mechanisms in consumer decision-making, echoing the findings of Andersen et al. (2009). Meanwhile, clarity in presenting sustainability comparisons significantly enhanced EWOM across both countries, suggesting that comprehensible impact data plays a critical role in fostering engagement. These findings contribute to the literature by illustrating how blockchain-enabled sustainability information interacts with cultural trust mechanisms, offering a refined perspective on technology-driven sustainability communication (La et al., 2008; Kumar et al., 2022; Utz et al., 2022).

Third, our findings enrich the literature on blockchain in consumer sustainability communication by extending the function of blockchain affordances beyond verification and traceability. (Cozzio et al., 2023; Nygaard and Silkoset, 2022). Our results reveal that blockchain affordances can act as a trust-enhancing mechanism, indirectly influencing purchase intention through increased CT and RC. Such trust mechanisms are especially critical in digital commerce, where misinformation can trigger irrational consumer behavior as demonstrated in recent e-commerce contexts (Sarraf et al., 2024). These results enhance existing scholarship by showing how blockchain-based sustainability disclosures engage with culturally embedded trust dynamics, providing a more nuanced understanding of how technology facilitates sustainability communication.

Finally, our results extend earlier work on sustainable consumption, which has highlighted the persistence of the attitude–behavior gap (White et al., 2019) by clarifying the psychological mechanisms through which technology supports sustainable action. One of the key contributions of this study lies in uncovering that although BE-EWOM does not have a direct effect on CT, it does influence PI and RC, but this effect is observed exclusively within the Vietnamese sample. This reinforces the importance of cultural context, indicating that cultural norms and expectations surrounding online reviews moderate the influence of BE-EWOM on downstream consumer behaviors. These results align with Kushwaha et al. (2022), who demonstrated that online communities often exhibit herd-like dynamics and polarization unless anchored in verifiable and trusted information, underscoring the need for trust-enhancing mechanisms like blockchain in EWOM contexts. In addition, the differing influence of PSV on both PI and RC across the two countries underscores the importance of adopting a multi-dimensional understanding of sustainability value. When it comes to sustainability, French consumers show a huge "attitude-behavior gap" than Vietnamese consumers. This indicates a major cultural difference in how sustainable values are turned into purchase intention.

### 6.2. Practical implications

This research offers actionable guidance for luxury brands, resale platforms, and regulatory bodies seeking to strategically deploy blockchain in sustainability-driven fashion markets. Crucially, although blockchain improves immutability, clarity, and transparency, its behavioral influence varies by market. Implementing blockchain in luxury resale markets is neither immediate nor effortless. It requires substantial upfront investment, adequate technical infrastructure, and coordinated adoption across often-fragmented supply networks. Therefore, blockchain should be approached as a gradual capability-building process rather than as a simple communication enhancement.

For Vietnamese consumers, blockchain is most impactful when positioned as an interactive provenance mechanism that facilitates peer-to-peer authentication and encourages community engagement. Firms should therefore design blockchain interfaces that enable consumers to explore sustainability histories directly, rather than treating blockchain merely as a backend traceability tool. Since electronic word-of-mouth (EWOM) functions as a powerful social validation loop in Vietnam's digital environment, platforms should actively foster community co-validation features, such as blockchain-verified consumer reviews or collaborative interpretation of provenance information. This approach leverages Vietnam's collectivist culture and rapidly expanding digital commerce ecosystem, allowing blockchain to serve not only as a certification mechanism but also as a catalyst of shared sustainability knowledge and trusted peer influence. Additionally, perceived sustainability value (PSV) significantly influences purchasing behavior in Vietnam. Blockchain-enabled sustainability impact comparisons, such as showcasing carbon or water savings, can significantly boost EWOM and trust, especially in Vietnam with a stronger effect. To maximize engagement, luxury brands should integrate AI-driven impact comparison tools into blockchain platforms and employ visualization to help explain their sustainability benefits. Marketing efforts should frame sustainability as a value tied to national pride (e.g., "If you buy second-hand luxury items in Vietnam, it equals helping us to go green").

For French consumers, our findings indicate a stronger preference for clarity and institutional legitimacy rather than community-driven authentication. Accordingly, blockchain-enabled provenance information should be incorporated into existing eco-label and third-party certification systems, rather than promoted as a standalone technological alternative. Communication strategies should minimize technical opacity by translating blockchain-derived indicators (e.g., carbon reduction metrics or certification hashes) into concise sustainability scorecards or interpretive summaries. Because customer trust in France is anchored in regulatory credibility and historically established labeling frameworks, firms will benefit more from positioning blockchain as an enhancement to recognized certification schemes than as an independent mechanism for verifying sustainability. While PSV influences trust, it does not directly drive behavior in France, indicating a greater "attitude–behavior gap." Thus, blockchain's clarity function should emphasize credibility and reinforce existing institutional cues to support consumer adoption.

For global luxury brands operating in both Vietnam and France, the observed differences do not necessarily imply a need for entirely separate blockchain platforms. Instead, a unified technical infrastructure could be combined with locally adapted interfaces and narratives that align with each market's dominant trust formation mechanisms. For example, the same blockchain-verified record may be presented through community-based co-validation features in Vietnam, but integrated into institutional eco-label systems and concise sustainability scorecards in France.

Finally, from a policy and loyalty strategy perspective, relationship commitment (RC) is a key determinant in driving purchase intent in both markets, with a stronger impact in France, underscoring the importance of long-term consumer engagement strategies. For luxury brands, blockchain-powered loyalty programs should be capitalized to enhance commitment, such as tokenized rewards (e.g., NFTs certifying sustainability contributions) that create exclusivity. In Vietnam, gamified experiences connected to blockchains can be taken into consideration so that consumers repeat their purchases of sustainable products. Whereas in France, luxury brands can use blockchain to create VIP experiences such as exclusive early access for customers to limited-edition collections of clothes made from all sorts of recycled materials to strengthen customer loyalty. Policymakers should establish harmonized, product-level sustainability indicators for blockchain-enabled resale platforms, such as verified carbon footprint reductions, water usage savings, and product longevity metrics, and mandate their disclosure through blockchain-based digital product passports to enable transparent, auditable, and comparable sustainability assessments across

platforms. In France, regulatory mandates requiring blockchain verification for resale luxury goods can further strengthen customer trust and perceptions of sustainability. At the same time, the adoption of engagement strategies built on blockchains can be further pushed by providing tax incentives while making sure that they comply with data privacy regulations, including GDPR in the EU and PDPA in Vietnam. Ultimately, blockchain's role in sustainability marketing hinges on market-specific strategies that align with cultural trust factors and regulatory environments, allowing brands to build trust, drive sustainable consumption, and foster long-term consumer relationships.

## 7. Limitations and future research

While this research has limitations, it also provides directions for future research. First, methodological constraints include the cross-sectional design, which limits the ability to infer causality. Longitudinal or experimental research designs could help track how blockchain adoption influences customer trust and behavior over time. Additionally, the use of a self-developed survey instrument may introduce bias related to respondent self-selection, recall, or social desirability. Mixed-methods approaches incorporating qualitative interviews or behavioral experiments could offer richer, more triangulated insights. In addition, data were collected from consumers who are familiar with blockchain technology, which may introduce a further layer of self-selection bias. Moreover, because recruitment was conducted primarily through online and social media channels, the sample may further overrepresent digitally engaged consumers relative to less tech-savvy luxury consumers. Respondents with prior blockchain familiarity are likely to exhibit higher technological readiness and greater openness toward blockchain-enabled sustainability information than the general consumer population. Future research could address this limitation by adopting broader and more systematic sampling strategies, such as stratified sampling designs that explicitly include consumers with varying levels of blockchain familiarity and digital engagement. Although we observed no systematic patterns of missingness, this study did not conduct a detailed analysis of missing data mechanisms, which should be considered a minor limitation. Future research could apply more rigorous missing data diagnostics and imputation techniques to further strengthen the robustness of empirical findings. This study relies on the linear modeling assumptions inherent in PLS-SEM. While this approach is widely used in consumer behavior research, it may not fully capture potential nonlinear relationships among constructs. Future research could explore nonlinear modeling techniques or alternative analytical approaches to examine whether such relationships offer additional explanatory power.

Second, although we interpret the Vietnam–France differences partly through country-level factors such as technology readiness and institutional safeguards, these macro-level conditions were not directly measured or modeled in this study. We also did not directly capture institutional legitimacy (e.g., trust in certification/regulatory regimes or platform governance) or shared credibility norms. Future research should apply multigroup or multilevel models that formally test these contextual moderators to confirm the robustness of cultural interpretations.

Third, the study focuses on national differences but does not examine intergenerational variation in consumer behavior. With Generation Alpha and younger digital-native cohorts increasingly influencing consumption dynamics, future research could segment respondents by age or digital literacy to capture more fine-grained behavioral insights. Relatedly, our sample likely reflects a more digitally active segment of luxury resale consumers; therefore, generalisation to less tech-savvy consumers or lower-involvement segments should be made with caution.

Fourth, although the study focuses on the second-hand luxury segment, "luxury" was operationalised using a category-based definition to ensure cross-country comparability. This approach may mask within-country heterogeneity in luxury perceptions, for example, differences in brand knowledge, luxury involvement, income constraints, or distinctions between entry-level and ultra-luxury consumption. Future research could incorporate price-based thresholds or perceptual measures of luxury and test within-country heterogeneity using multigroup, latent class, or moderation analyses.

Fifth, while this study investigates blockchain-enabled sustainability information, it does not account for newer technological innovations such as non-fungible tokens (NFTs), decentralized finance (DeFi), and smart contracts. These developments may reshape trust and consumer behavior, especially in digitally advanced or asset-driven consumption markets. Future studies should explore how these emerging tools interact with sustainability cues in purchase decisions.

Sixth, although MICOM confirms measurement invariance and allows meaningful comparison of structural relationships, the Vietnamese and French samples are not demographically matched in terms of income and occupational composition. These differences may partially confound cross-national comparisons, as demographic characteristics can shape technology readiness and sustainability sensitivity. Therefore, the observed cross-country differences should be interpreted as contextual differences within tech-aware luxury consumer segments, rather than as representative national-level cultural effects.

Finally, while this study identifies the influence of blockchain features on customer trust and purchase intention, it does not investigate how different communication strategies impact the perceived credibility of blockchain-based sustainability claims. Future research should examine message framing, platform design, or the role of third-party certifications in increasing consumer awareness and engagement, particularly in markets with varied levels of digital adoption.

## CRediT authorship contribution statement

**Khoi Minh Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Van Anh Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Chau Ho Diem Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Trang Thao Huynh:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Tran Phuong Dung:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **An Quach:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Data curation. **Ngan Thanh Nguyen:** Writing – review & editing, Writing – original draft, Project administration, Investigation.

## Ethical compliance

The research process involving human participants followed the ethical standards of the institutional and national research committees of the 1964 Helsinki Declaration and its subsequent amendments, as well as related ethical standards.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2026.104808.

## Data availability

Data will be made available on request.

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